

4DVarNet-FM: DirectUNet & VanillaCFM Benchmark Report

Experiment Configuration Summary

ID	Model	Channels	Emb	Epochs	Train Mix	Rand	Noise	N_outer	σ_{prior}
E1_direct_unet_default	DirectUNet	[64, 128, 256]	-	200	cs1+cs2	no	-	-	-
E2_direct_unet_small	DirectUNet	[32, 64, 128]	-	200	cs1+cs2	no	-	-	-
E3_direct_unet_rand	DirectUNet	[32, 64, 128]	-	200	cs1_rand+cs2_rand	yes	-	-	-
F1_vanilla_cfm_default	VanillaCFM	[64, 128, 256]	-	400	cs1+cs2	no	-	-	-
F2_vanilla_cfm_small	VanillaCFM	[32, 64, 128]	-	400	cs1+cs2	no	-	-	-
F3_vanilla_cfm_rand	VanillaCFM	[32, 64, 128]	-	400	cs1_rand+cs2_rand	yes	-	-	-

Shared settings (from lorenz63_default.yaml):
dt=0.01, T_max=3.0s, obs_interval=20, R_var=0.5, B_var=2.0
num windows=2000, spinup_steps=10000, batch_size=32
lr=0.001, gradient_clip_val=10.0 (stage1), stage2 disabled

CS1: param_bias=0.0, forcing_state_bias=0.0, coupling=linear (ideal conditions)
CS2: param_bias=0.15, forcing_state_bias=0.15, coupling=quartic (corrupted)
rand: randomize_params=true, param_noise=0.2 (uniform perturbation on σ , ρ , β)

DirectUNet: single UNet pass: obs -> state (MSE loss, no flow matching)
VanillaCFM: conditional flow matching with LinearInterpolant(nu=1.0)
Euler integration over N_outer steps, $x_0 \sim N(0, \sigma_{\text{prior}})$

Model Architectures

Model Architecture Details

Backbone: UNet1D (models/unet.py)

1D convolutional U-Net with down/up blocks

Input: state + observations (concatenated along channel dim)

Output: vector field v (for CFM) or state estimate (for DirectUNet)

DirectUNet (models/direct_unet.py):

- Wraps UNet1D with `time_emb_dim=0` (no time conditioning)
- `x0 = zeros(B, D, T)`, `tau = 0`, `out = unet(x0, obs, tau)`
- Loss: `MSE(out, states)` – direct regression
- No flow matching, no sampling loop
- Parameters: $\sim$ (hidden_channels dependent)

VanillaCFM (models/vanilla_cfm.py):

- Wraps UNet1D with `time_emb_dim=64` (sinusoidal time embedding)
- `LinearInterpolant(nu=1.0)`: `mix(x0, x1, t) = (1-t)*x0 + t*x1`
- CFM loss: `MSE(v_pred, v_target)` where `v_target = states - x0`
and `x0  $\sim$  N(0, sigma_prior=0.5)`
- Sampling: Euler integration over `N_outer=10` steps
`x_{t+1} = x_t + dt * v(x_t, obs, t)`
- Parameters: $\sim$ (hidden_channels dependent) + `time_emb`

Parameter counts (UNet1D backbone only):

E1/F1 [64,128,256]	encoder~123k	decoder~123k	total~246k
E2/E3/F2/F3 [32,64,128]	encoder~30k	decoder~30k	total~61k

Optimizer: Adam, `lr=0.001`, no scheduler

Gradient clipping: 10.0

Results Summary

Per-Variable Mean RMSE Summary

--- CS1 ---					
ID	Model	X	Y	Z	Mean
E1_direct_unet_default	DirectUNet	0.1549	0.1396	0.1379	0.1441
E2_direct_unet_small	DirectUNet	0.0474	0.0717	0.1253	0.0815
E3_direct_unet_rand	DirectUNet	0.1329	0.1346	0.0815	0.1163
F1_vanilla_cfm_default	VanillaCFM	0.1093	0.1534	0.1593	0.1407
F2_vanilla_cfm_small	VanillaCFM	0.0658	0.1329	0.1581	0.1190
F3_vanilla_cfm_rand	VanillaCFM	0.0495	0.0707	0.0868	0.0690
--- CS2 ---					
ID	Model	X	Y	Z	Mean
E1_direct_unet_default	DirectUNet	0.1728	0.1579	0.1383	0.1563
E2_direct_unet_small	DirectUNet	0.0731	0.0911	0.1418	0.1020
E3_direct_unet_rand	DirectUNet	0.1309	0.1301	0.0809	0.1139
F1_vanilla_cfm_default	VanillaCFM	0.1053	0.1665	0.1754	0.1491
F2_vanilla_cfm_small	VanillaCFM	0.0753	0.1501	0.1518	0.1257
F3_vanilla_cfm_rand	VanillaCFM	0.0514	0.0689	0.0895	0.0700

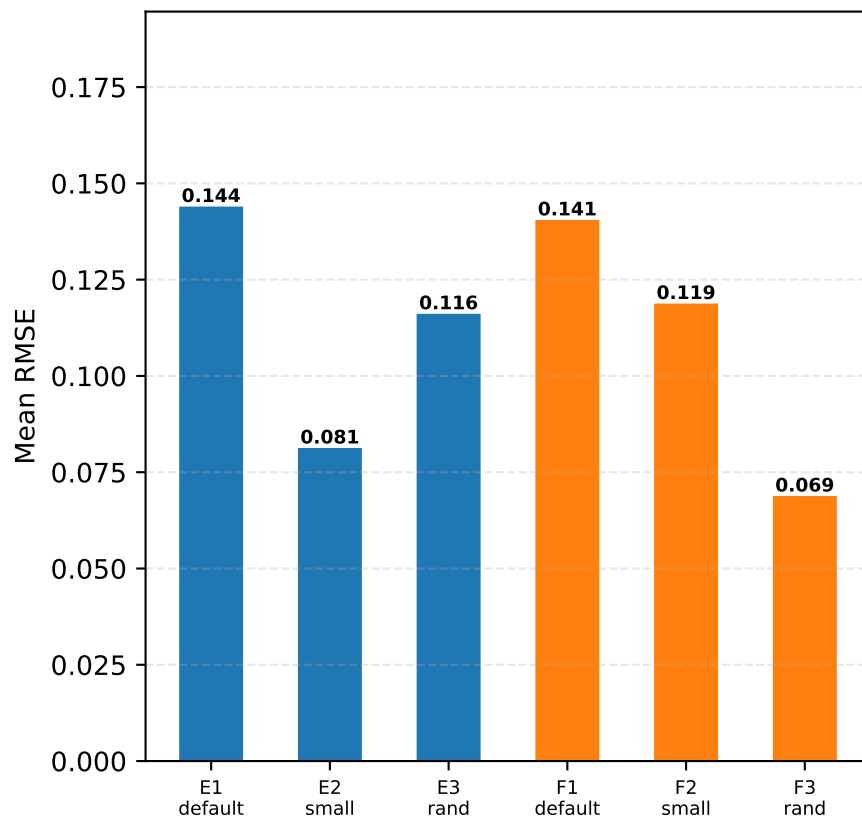
Degradation Analysis (CS2 μ / CS1 μ):

ID	Model	CS1 μ	CS2 μ	Deg	Time(s)	Std CS1	Std CS2	
E1_direct_unet_default	DirectUNet	0.1441	0.1563	1.08	x 482	0.0178	0.0183	
E2_direct_unet_small	DirectUNet	0.0815	0.1020	1.25	x 393	0.0044	0.0055	☆ GOOD
E3_direct_unet_rand	DirectUNet	0.1163	0.1139	0.98	x 398	0.0122	0.0127	
F1_vanilla_cfm_default	VanillaCFM	0.1407	0.1491	1.06	x 980	0.0206	0.0202	
F2_vanilla_cfm_small	VanillaCFM	0.1190	0.1257	1.06	x 721	0.0151	0.0152	
F3_vanilla_cfm_rand	VanillaCFM	0.0690	0.0700	1.01	x 789	0.0157	0.0154	★ BEST

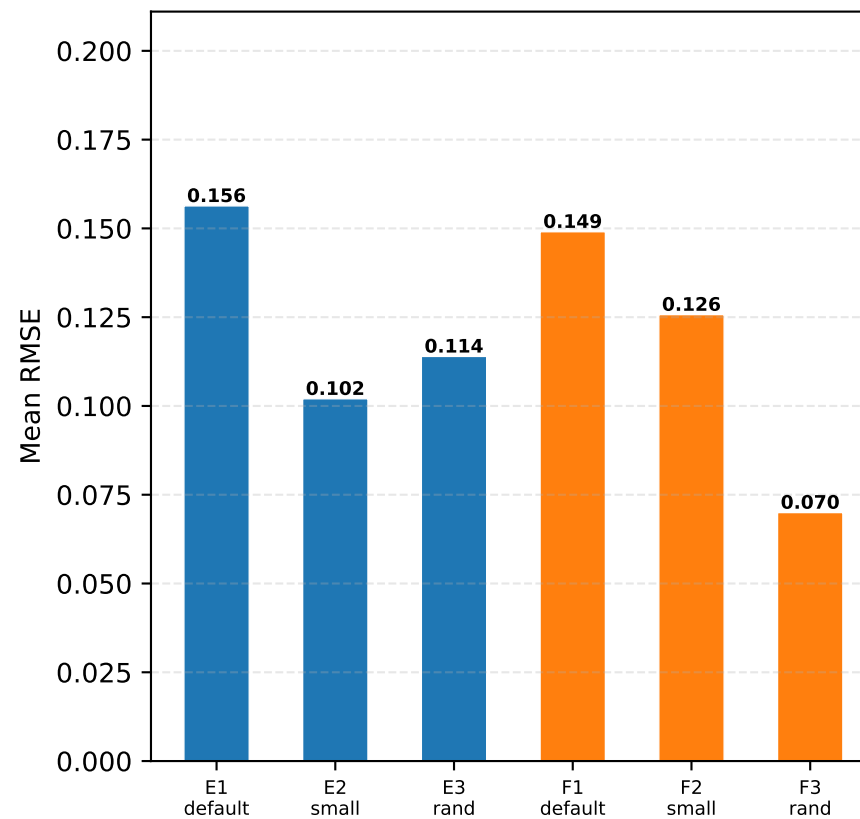
Interpretation:

Deg = CS2 μ / CS1 μ . Lower $\rightarrow$ more robust to perturbations.
Deg < 1.0 means model performs better on CS2 than CS1.
Best overall: F3 (VanillaCFM rand) – 0.069 CS1, 0.070 CS2, 1.014x deg

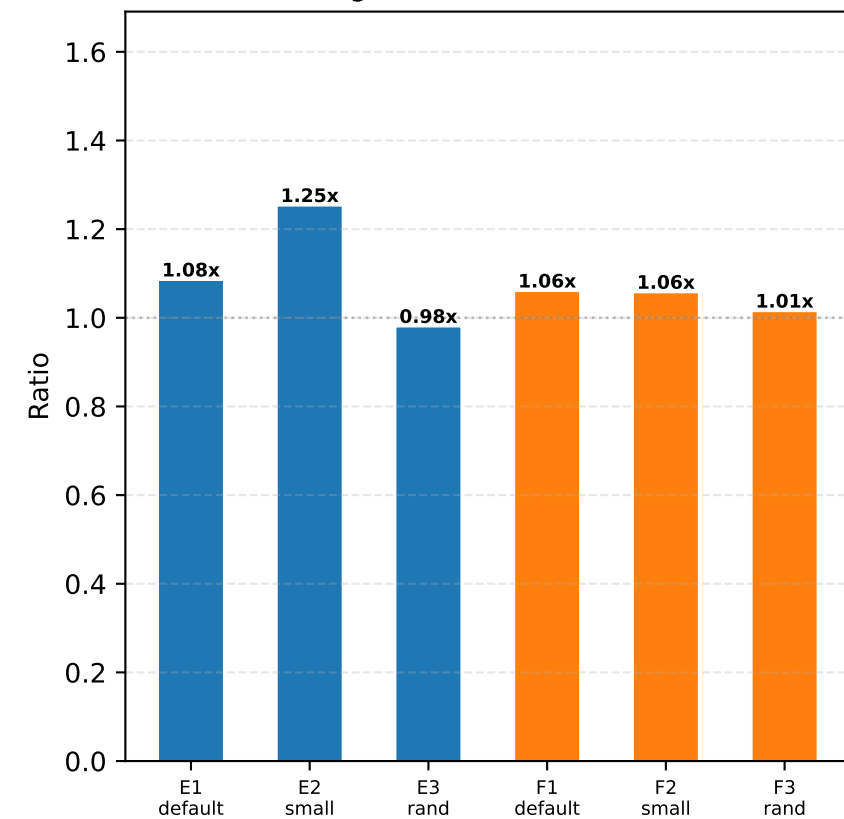
CS1 Mean RMSE



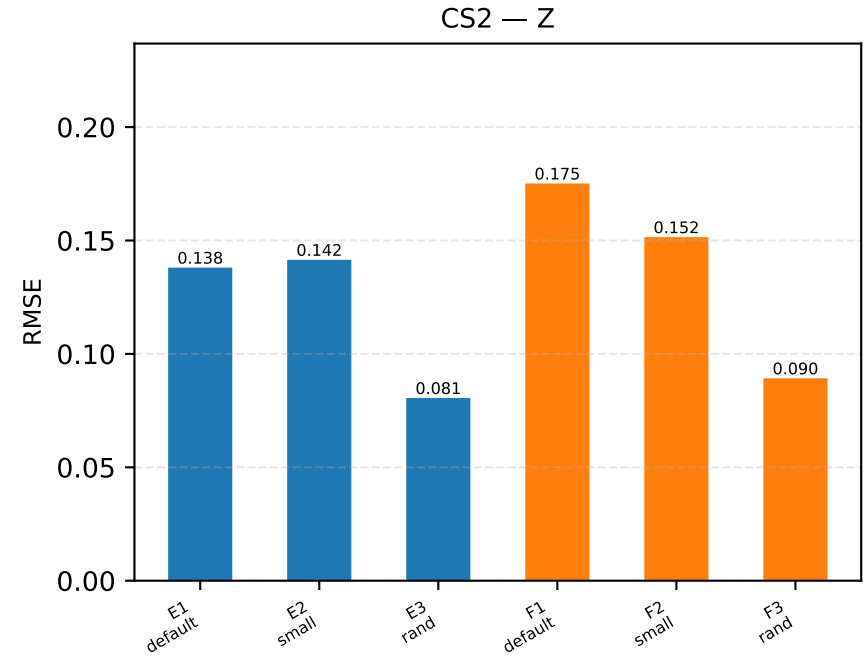
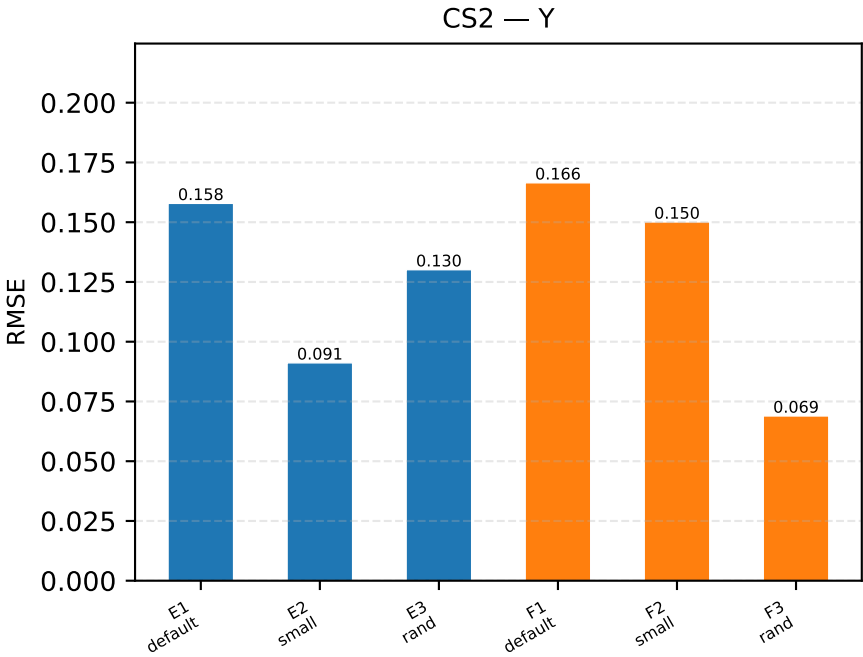
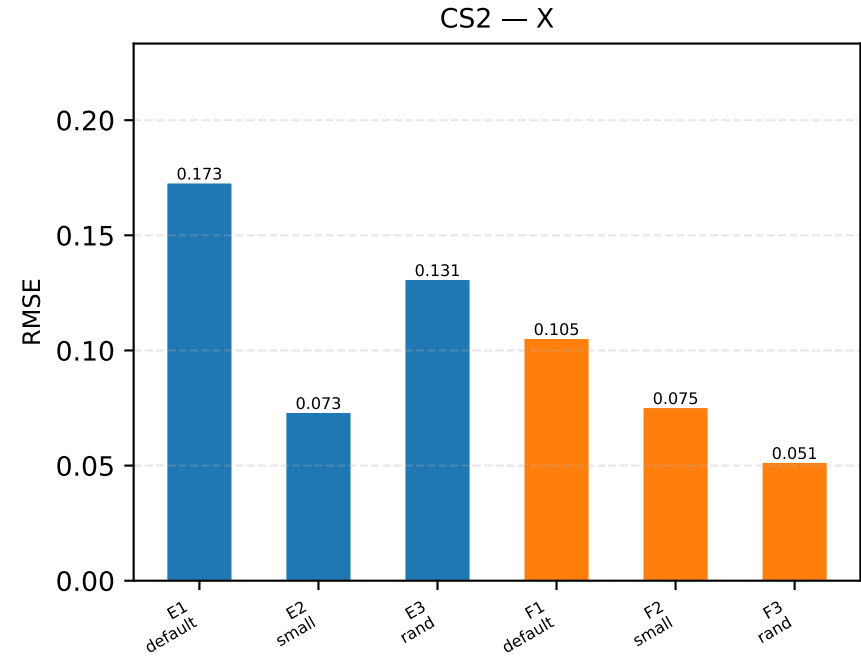
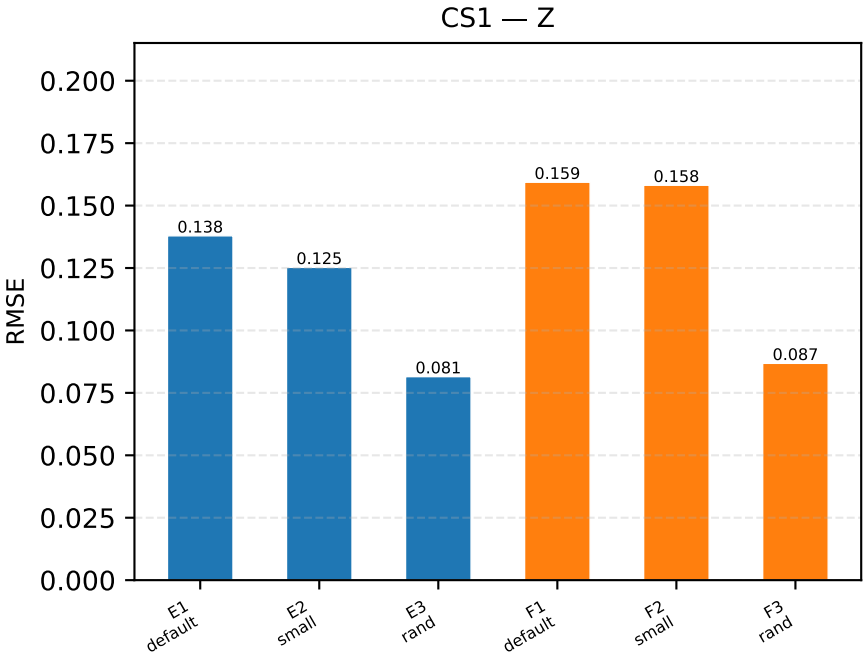
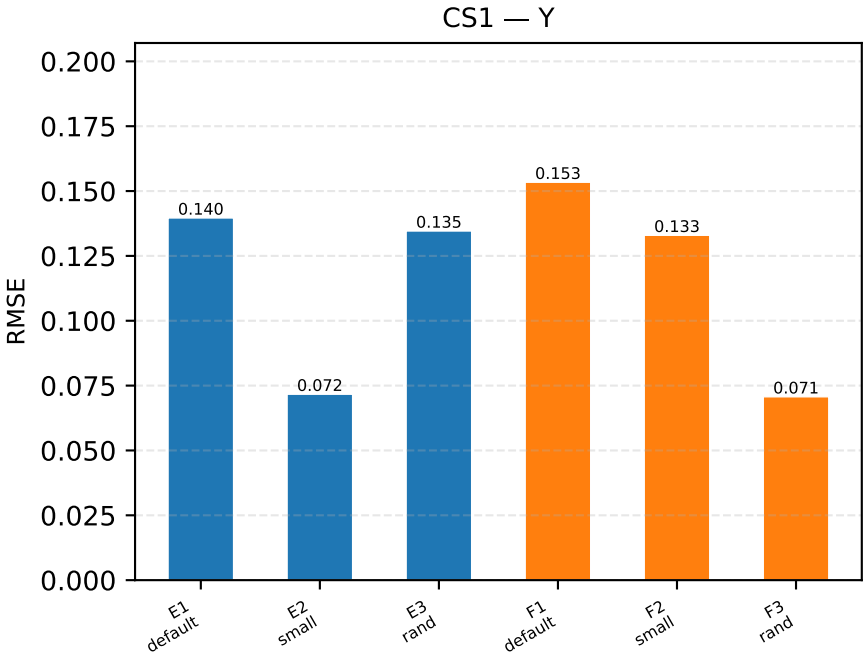
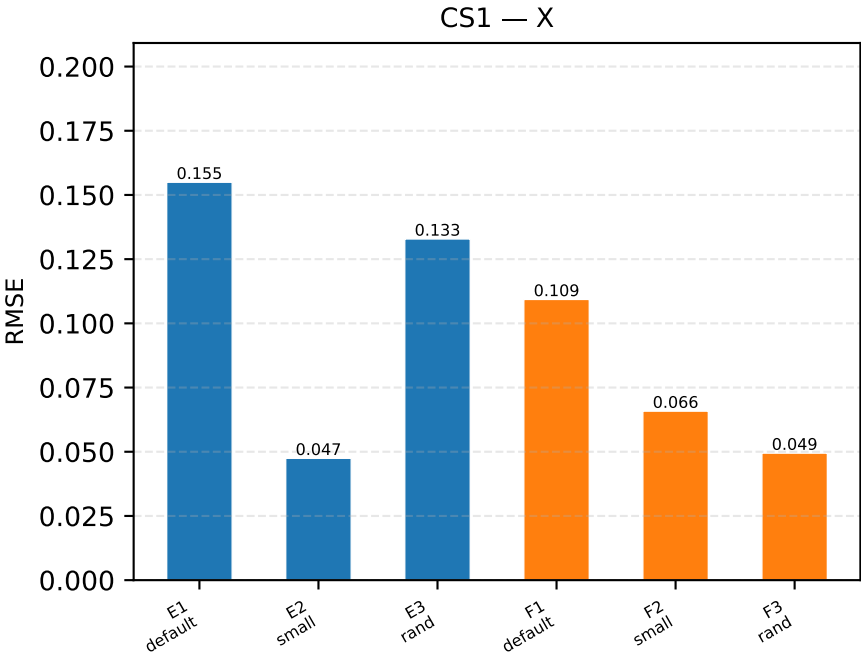
CS2 Mean RMSE



Degradation (CS2 / CS1)

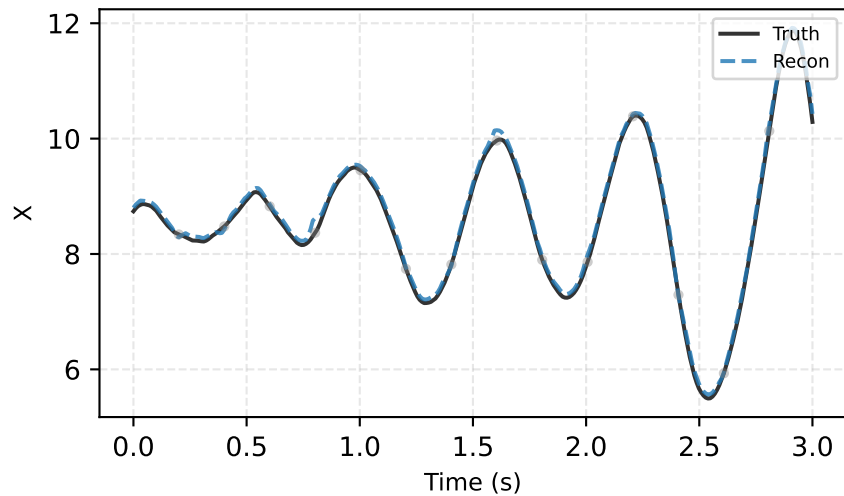


Per-Component RMSE Breakdown

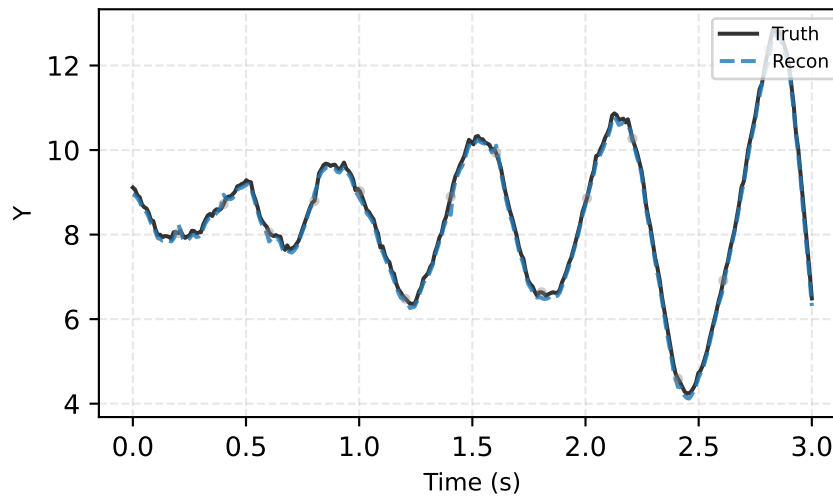


E1_direct_unet_default (DirectNet) - CSI Best & Worst Reconstruction

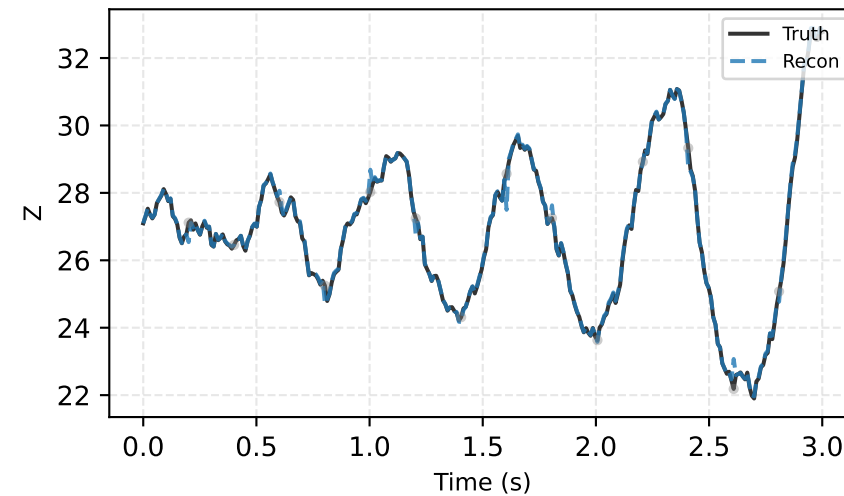
Best (traj #54) RMSE=0.101



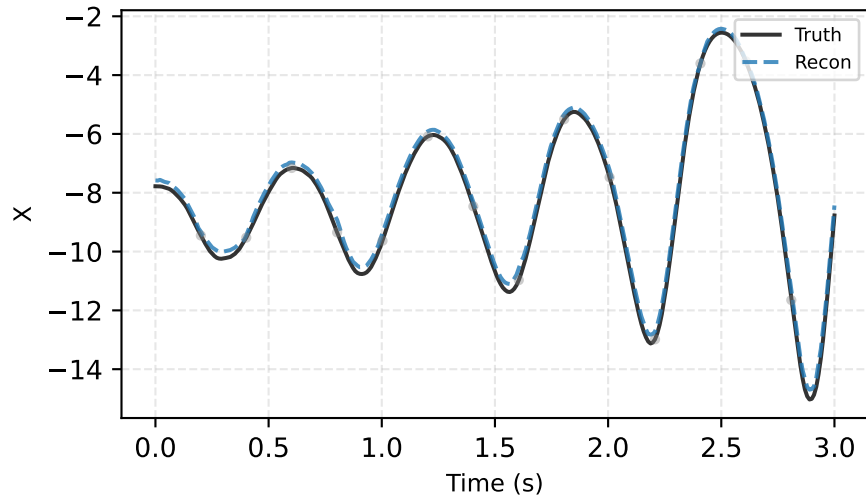
Best (traj #54) RMSE=0.101



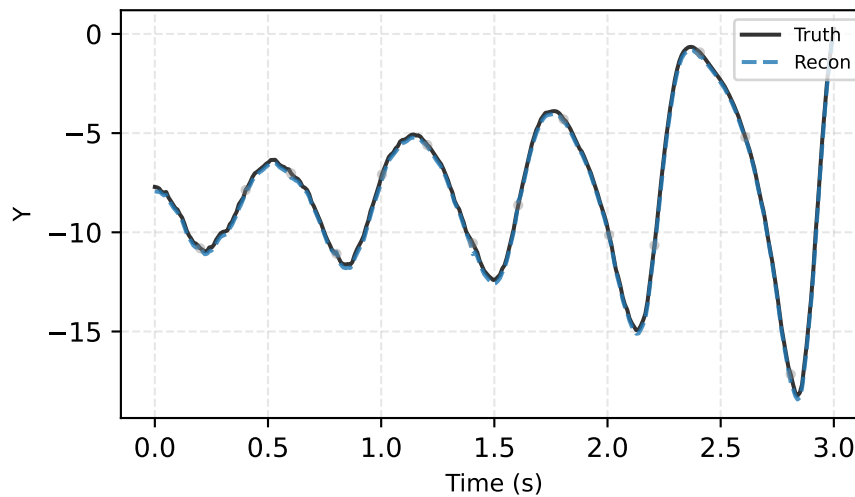
Best (traj #54) RMSE=0.101



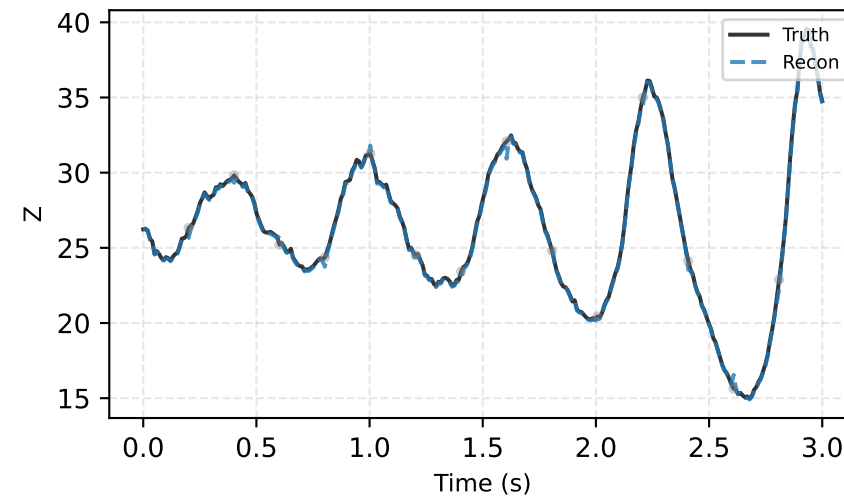
Worst (traj #24) RMSE=0.182



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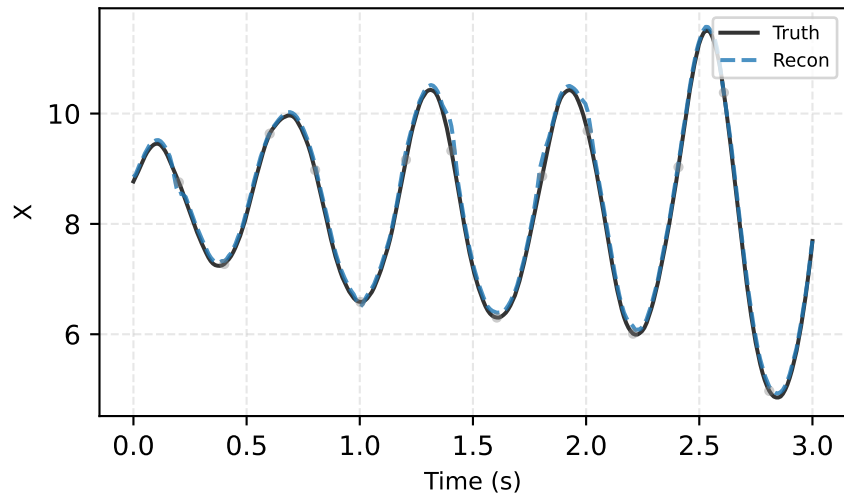


Worst (traj #24) RMSE=0.182

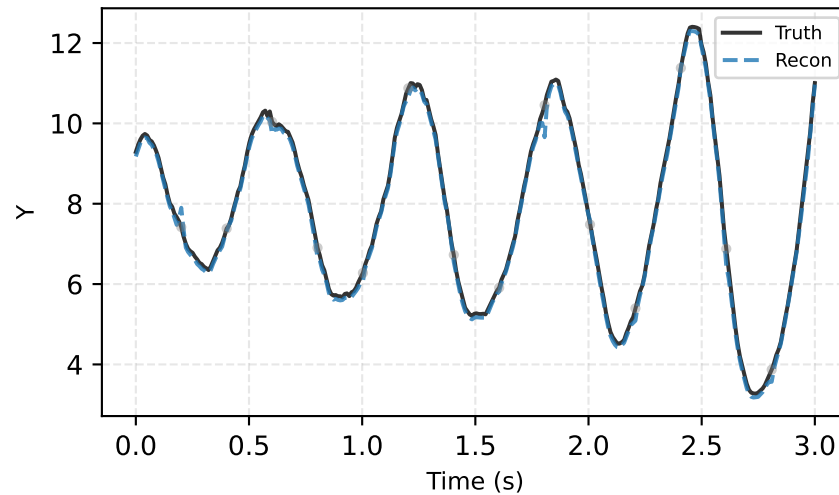


E1_direct_unet_default (DirectNet) — CS2 Best & Worst Reconstruction

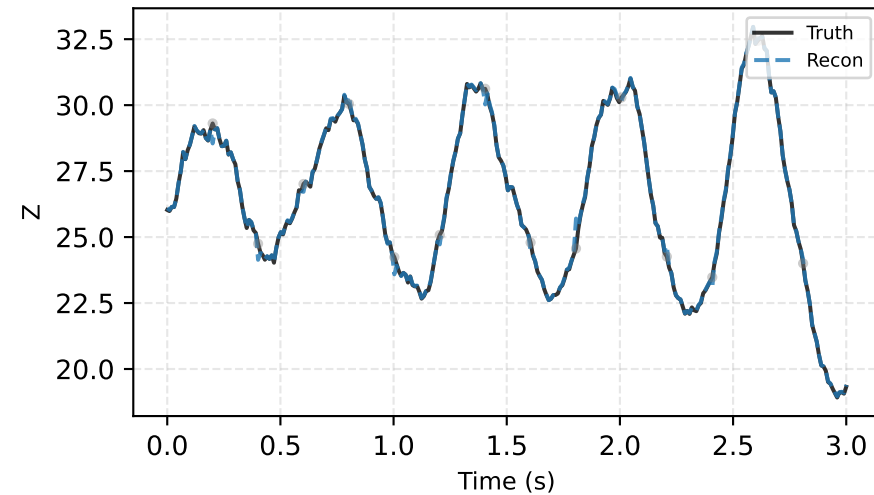
Best (traj #172) RMSE=0.113



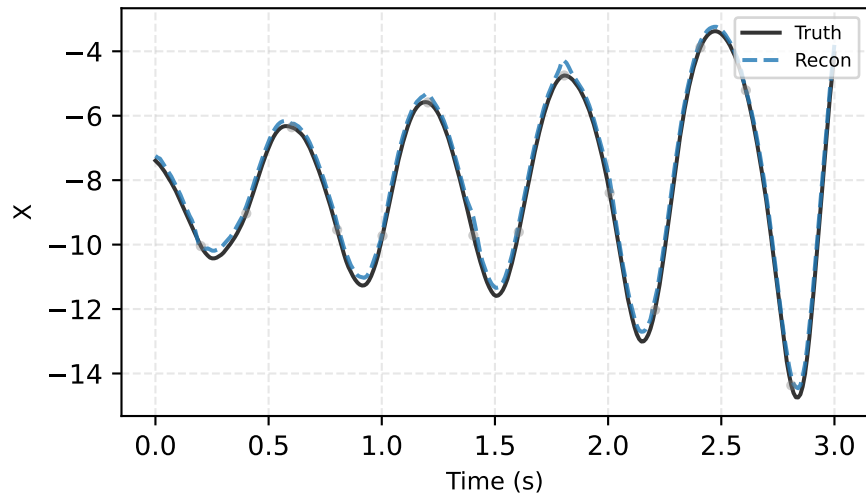
Best (traj #172) RMSE=0.113



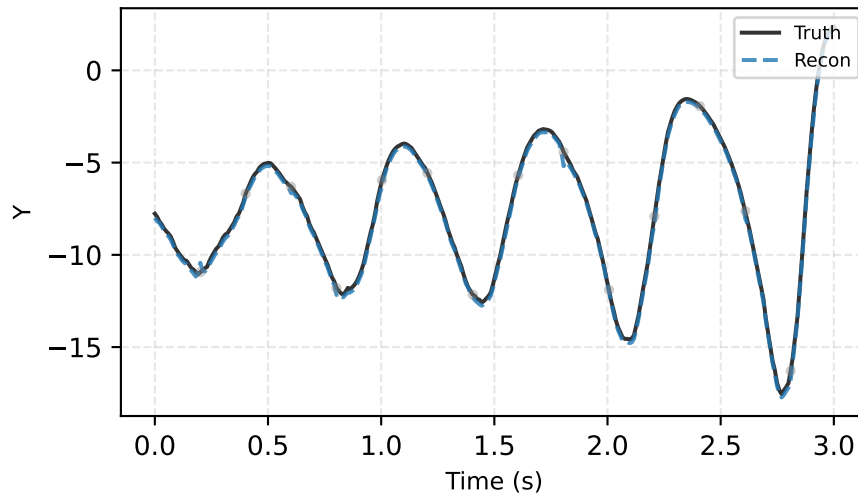
Best (traj #172) RMSE=0.113



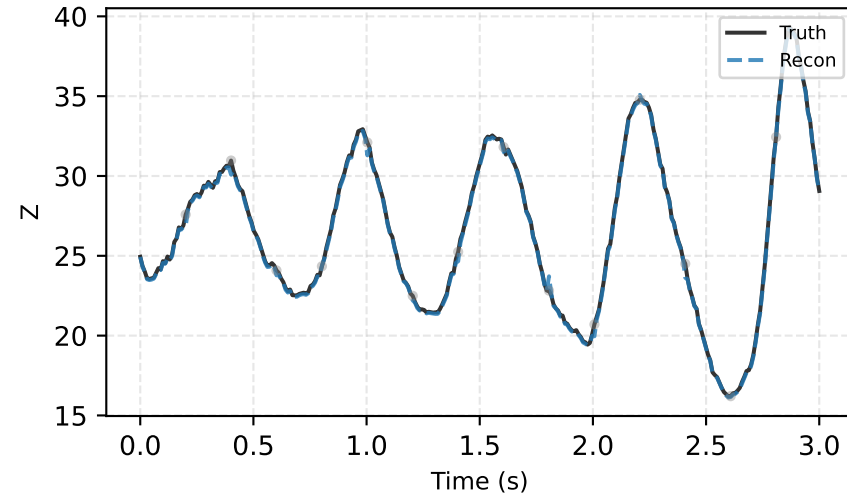
Worst (traj #174) RMSE=0.189



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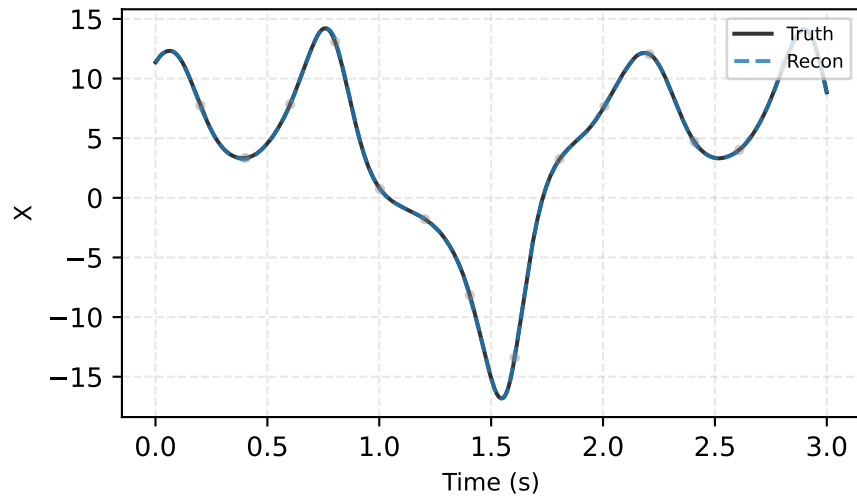


Worst (traj #174) RMSE=0.189

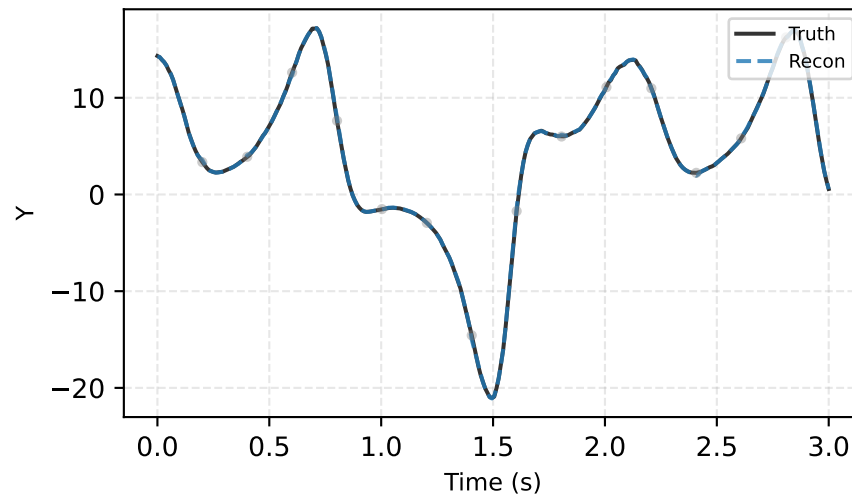


E2_direct_unet_small (DirectUNet) - CSI Best & Worst Reconstruction

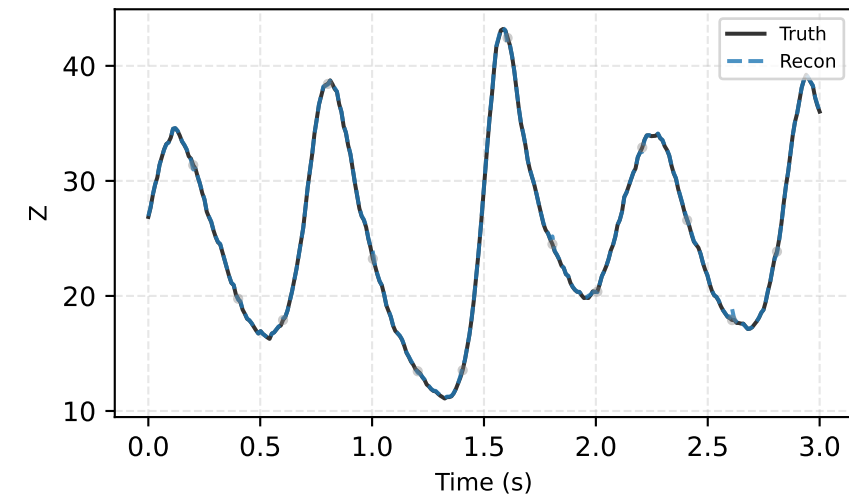
Best (traj #154) RMSE=0.078



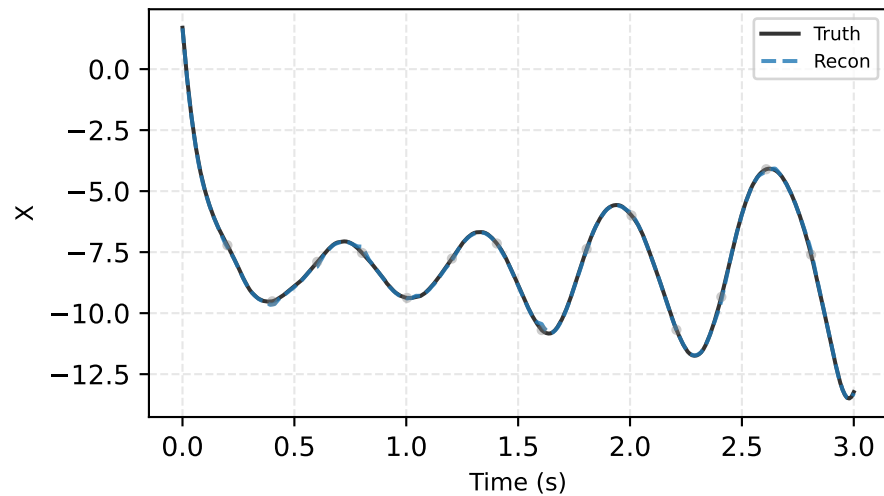
Best (traj #154) RMSE=0.078



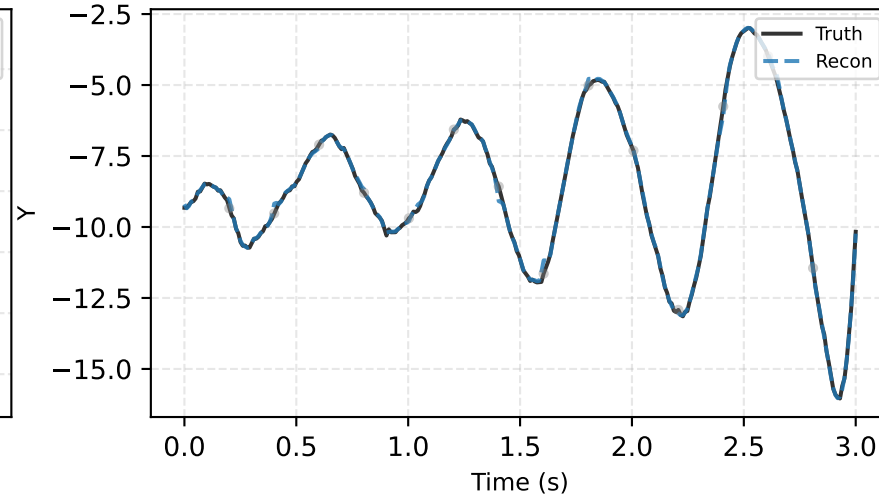
Best (traj #154) RMSE=0.078



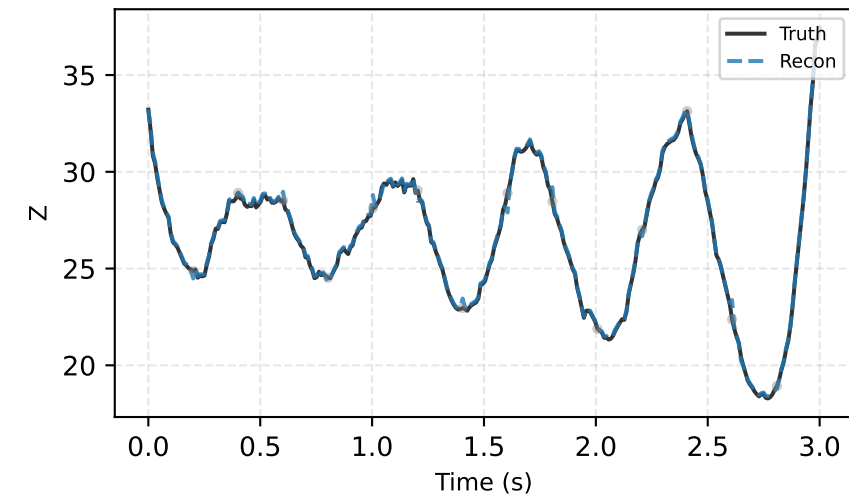
Worst (traj #102) RMSE=0.100



Worst (traj #102) RMSE=0.100

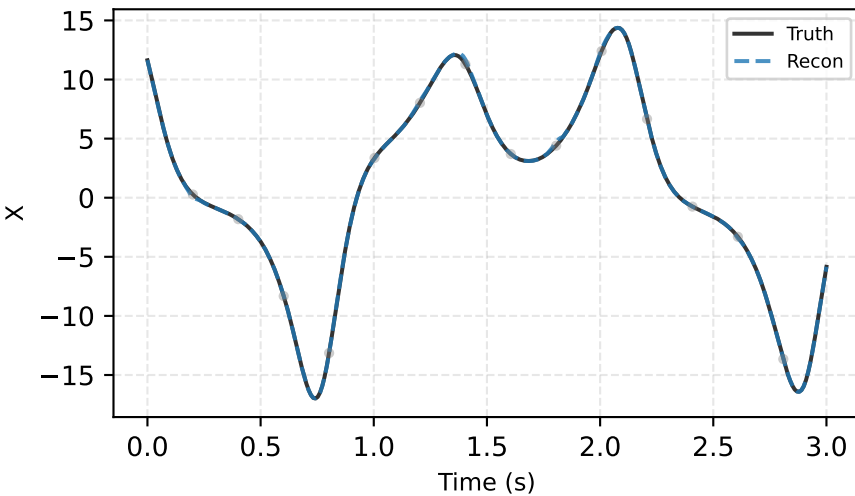


Worst (traj #102) RMSE=0.100

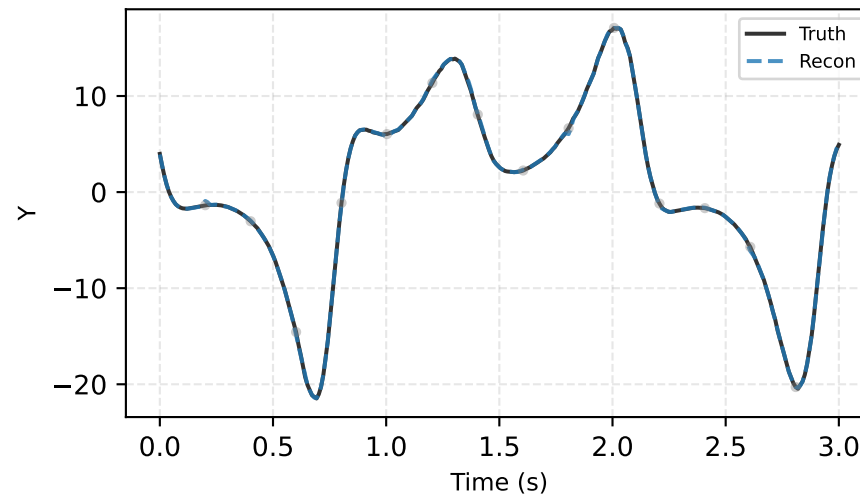


E2_direct_unet_small (DirectNet) — CS2 Best & worst Reconstruction

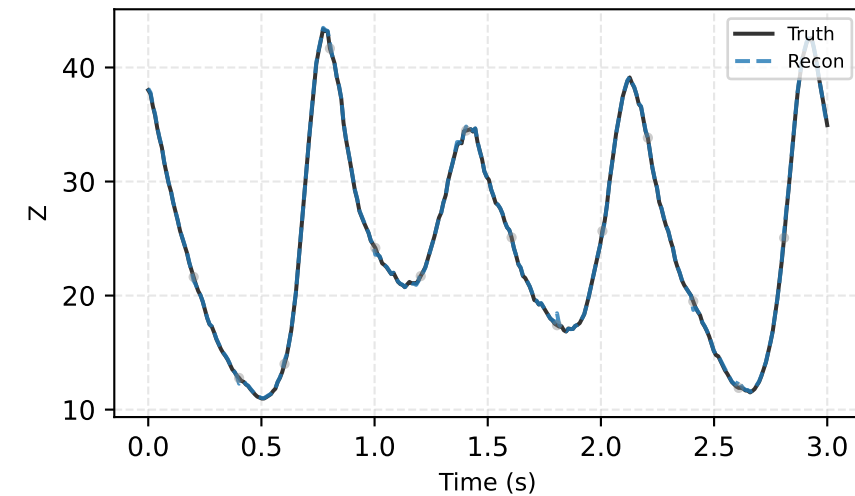
Best (traj #26) RMSE=0.098



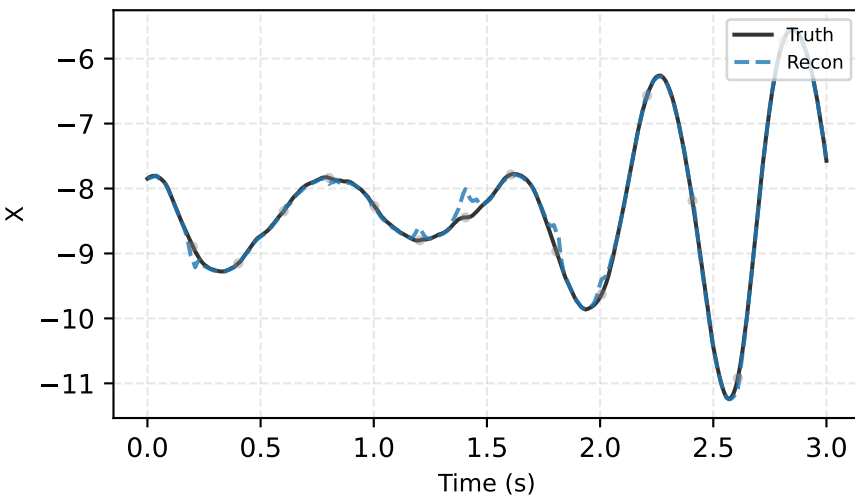
Best (traj #26) RMSE=0.098



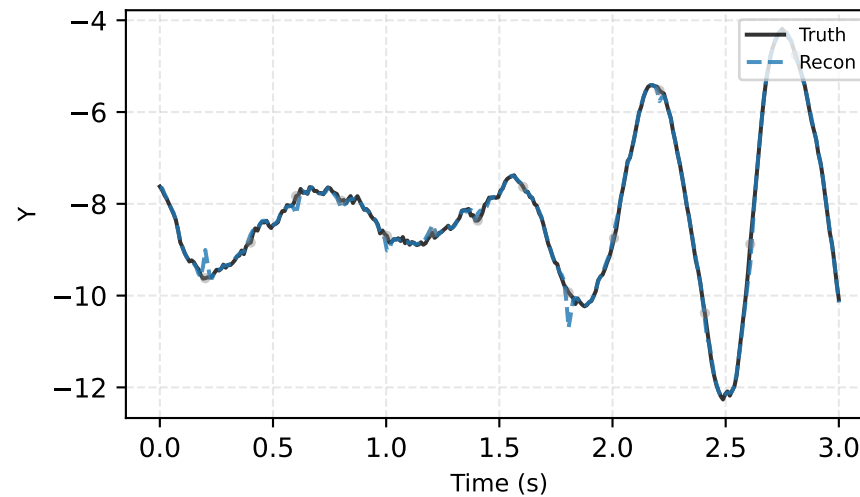
Best (traj #26) RMSE=0.098



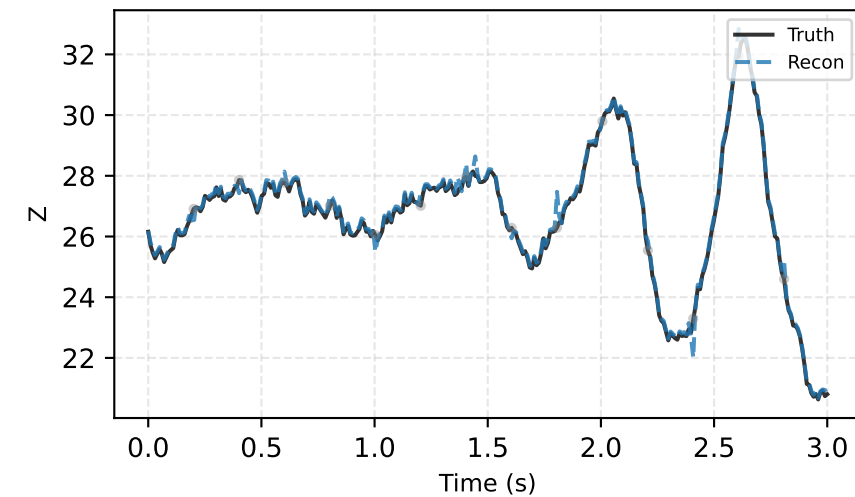
Worst (traj #149) RMSE=0.120



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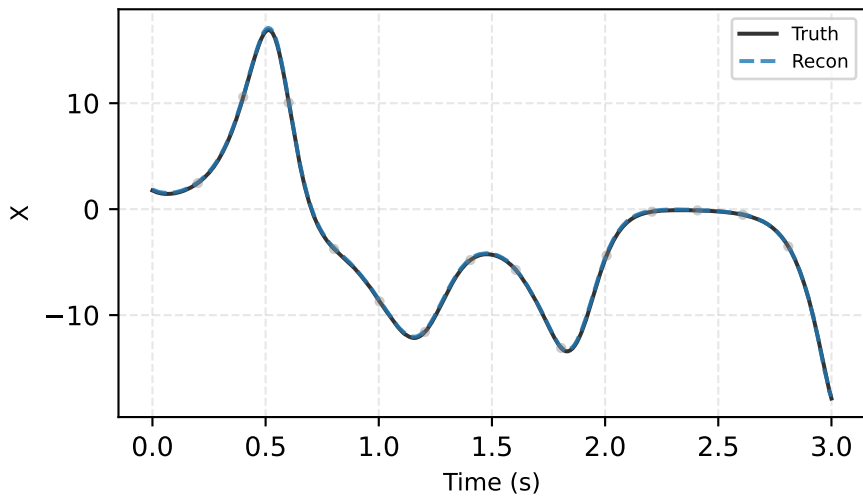


Worst (traj #149) RMSE=0.120

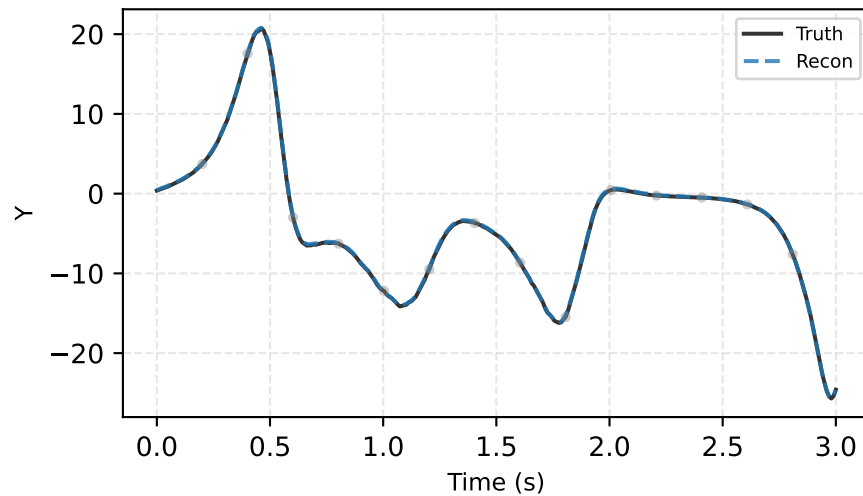


E5_direct_unet_rand (DirectNet) — CS1 Best & worst Reconstruction

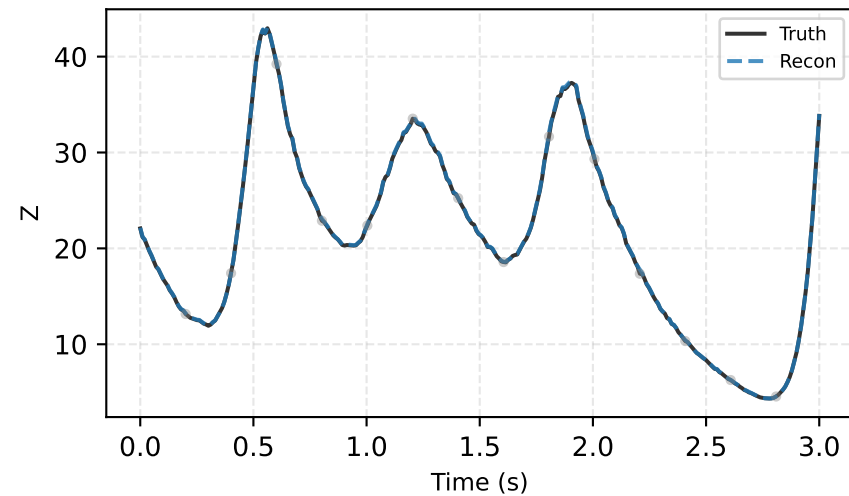
Best (traj #149) RMSE=0.097



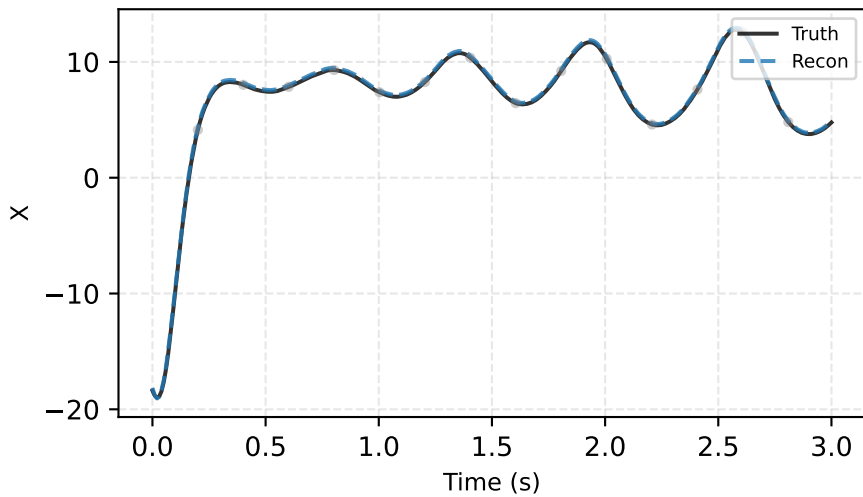
Best (traj #149) RMSE=0.097



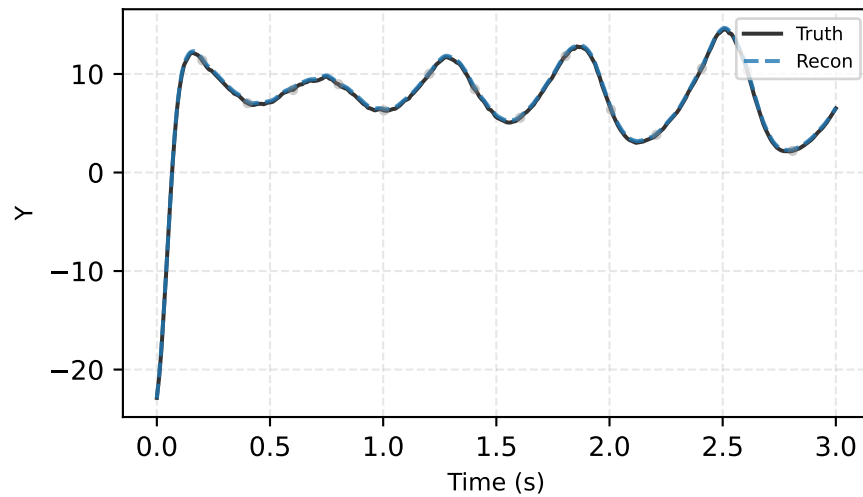
Best (traj #149) RMSE=0.097



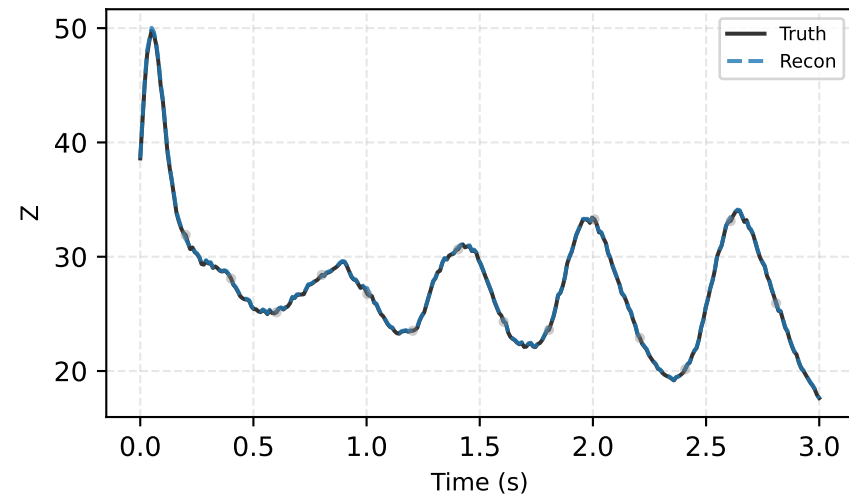
Worst (traj #80) RMSE=0.144



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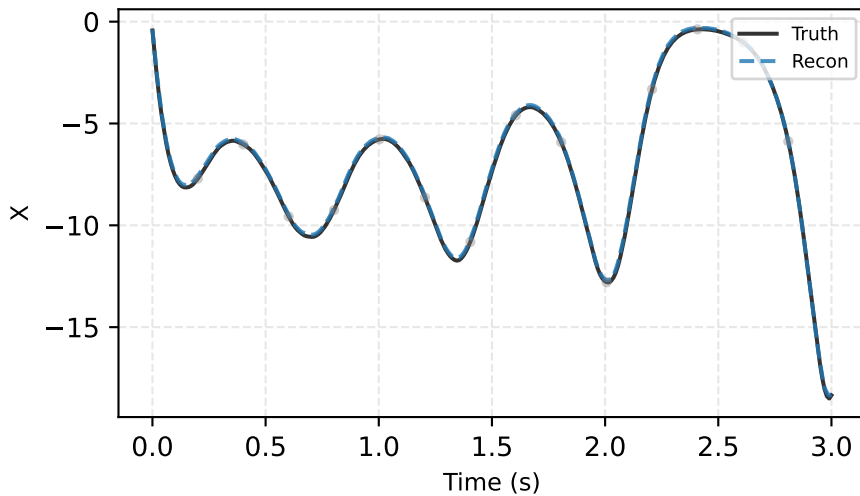


Worst (traj #80) RMSE=0.144

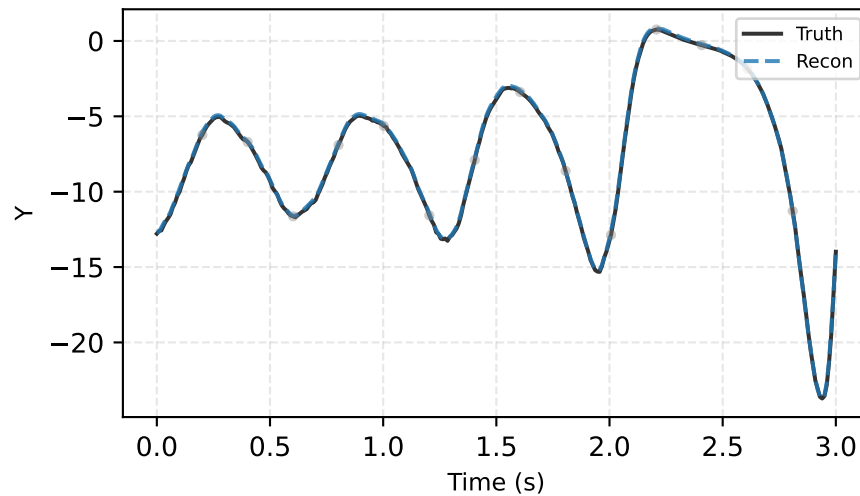


CS2_direct_unet_rand (DirectoNet) — CS2 Best & worst Reconstruction

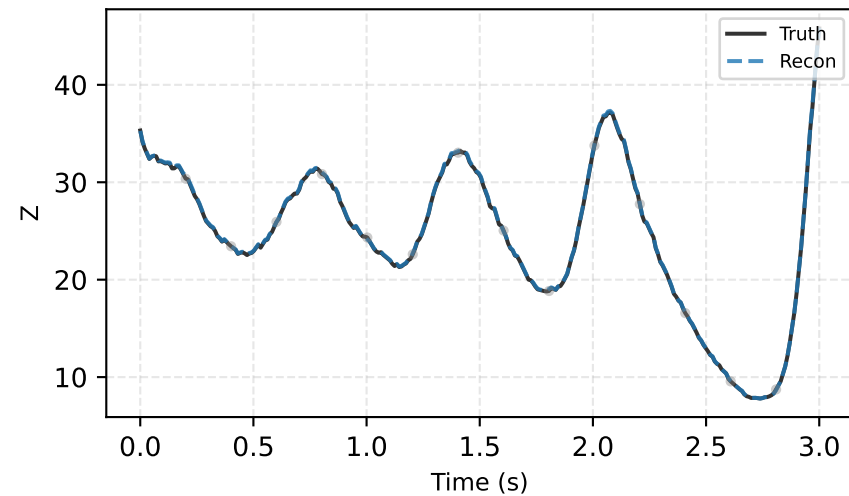
Best (traj #185) RMSE=0.099



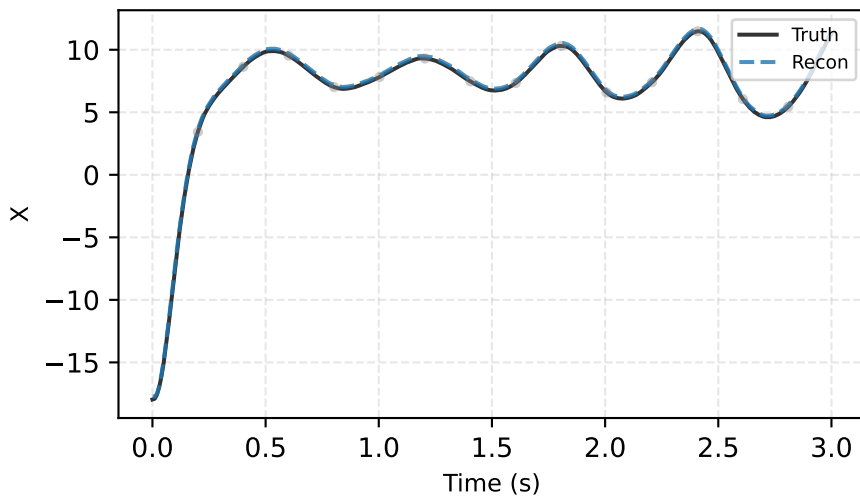
Best (traj #185) RMSE=0.099



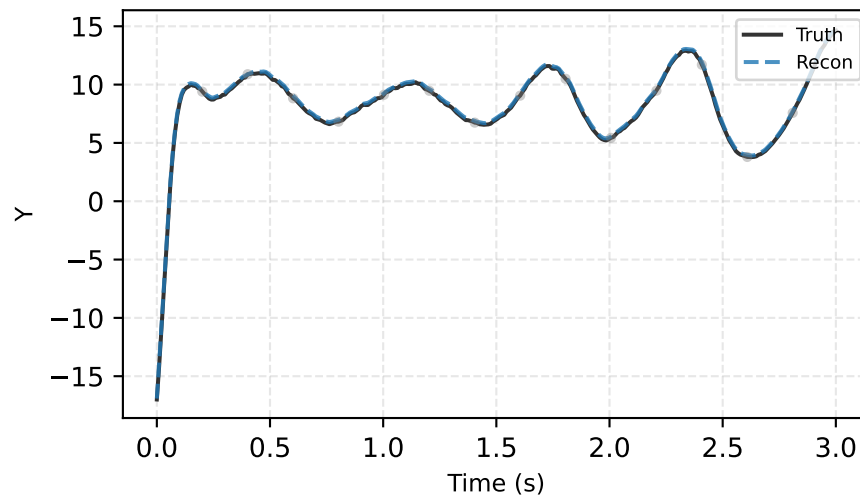
Best (traj #185) RMSE=0.099



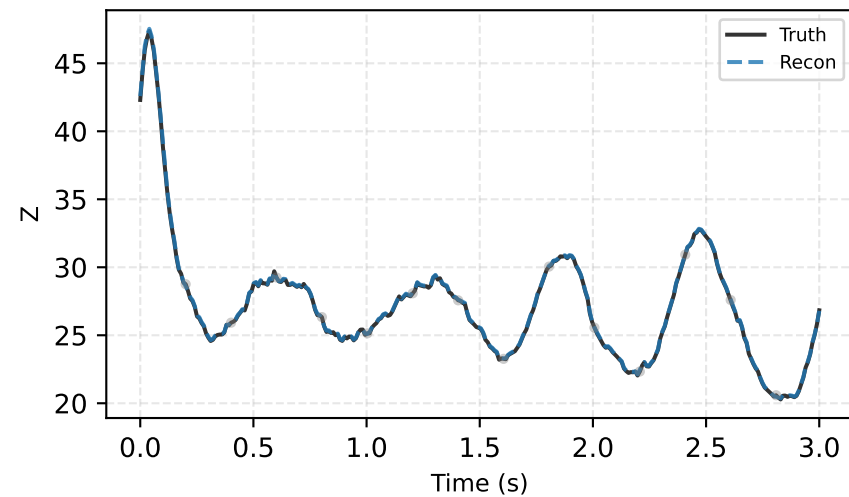
Worst (traj #127) RMSE=0.138



Worst (traj #127) RMSE=0.138

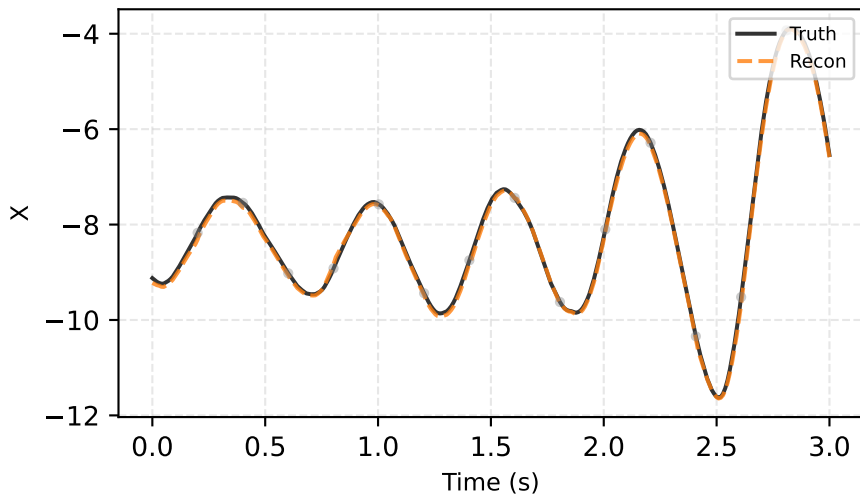


Worst (traj #127) RMSE=0.138

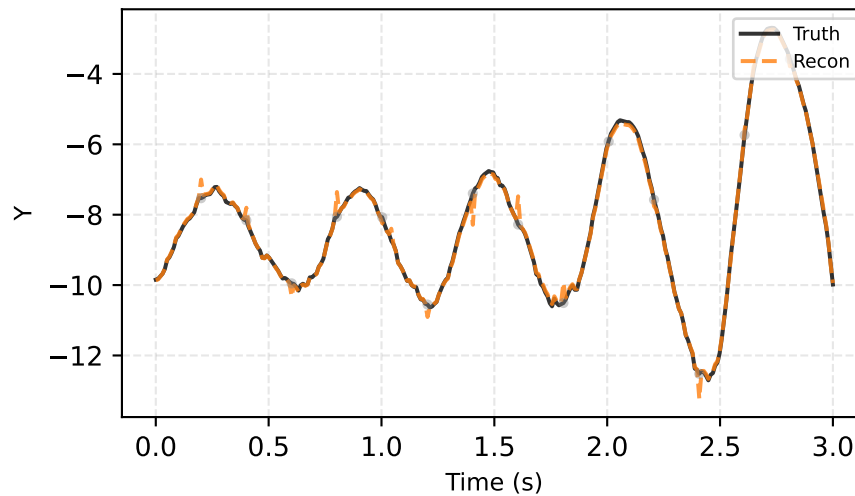


F1_vanilla_cnn_default (vanillaCFM) — CSI Best & worst Reconstruction

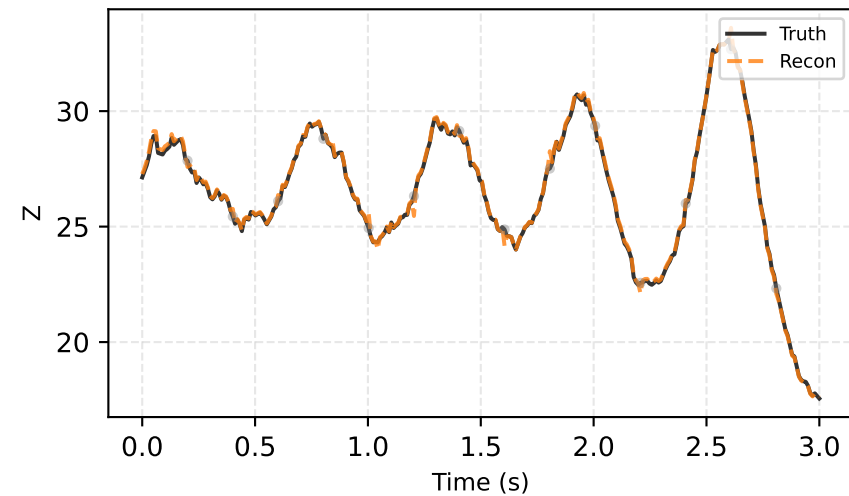
Best (traj #81) RMSE=0.110



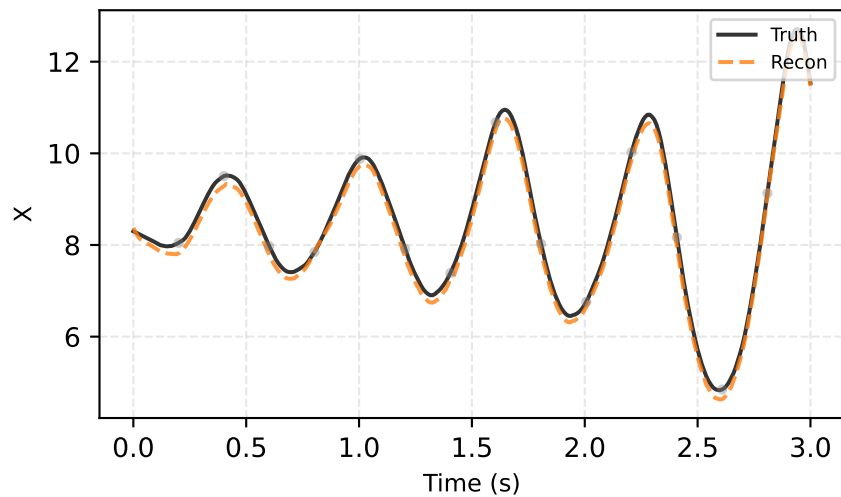
Best (traj #81) RMSE=0.110



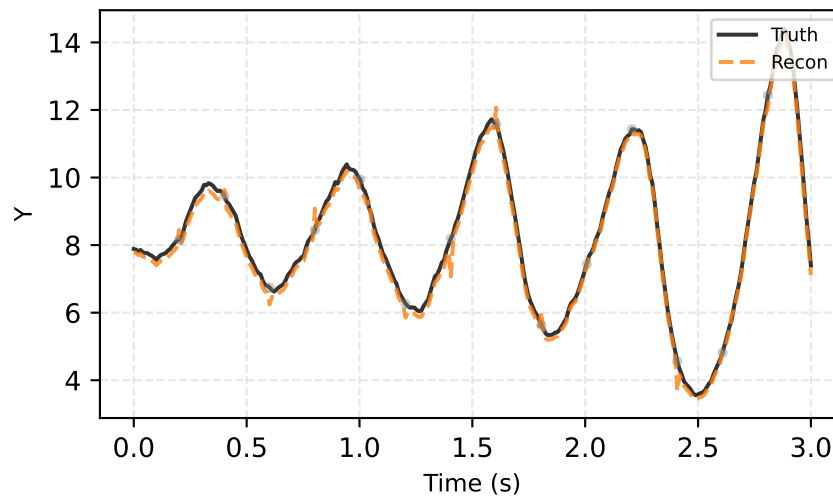
Best (traj #81) RMSE=0.110



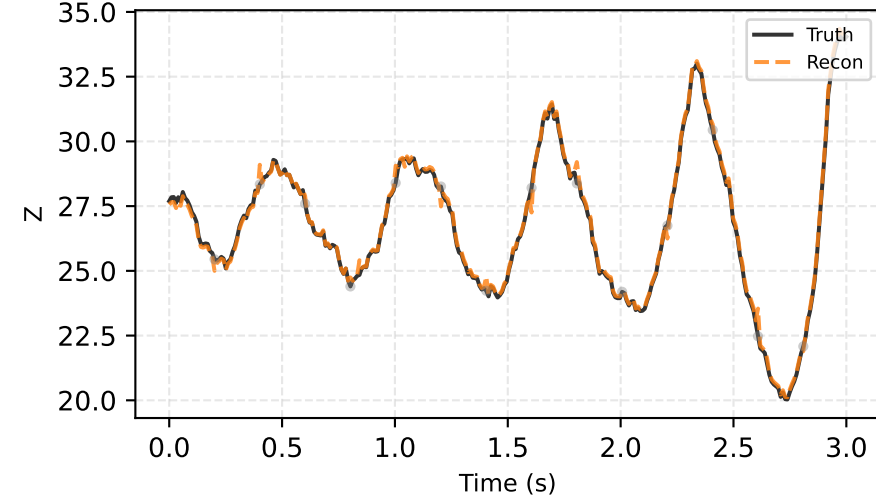
Worst (traj #22) RMSE=0.178



Worst (traj #22) RMSE=0.178

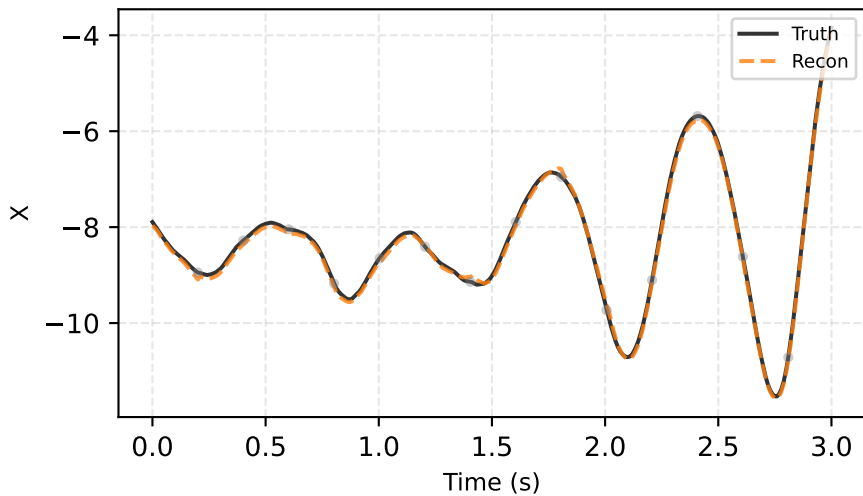


Worst (traj #22) RMSE=0.178

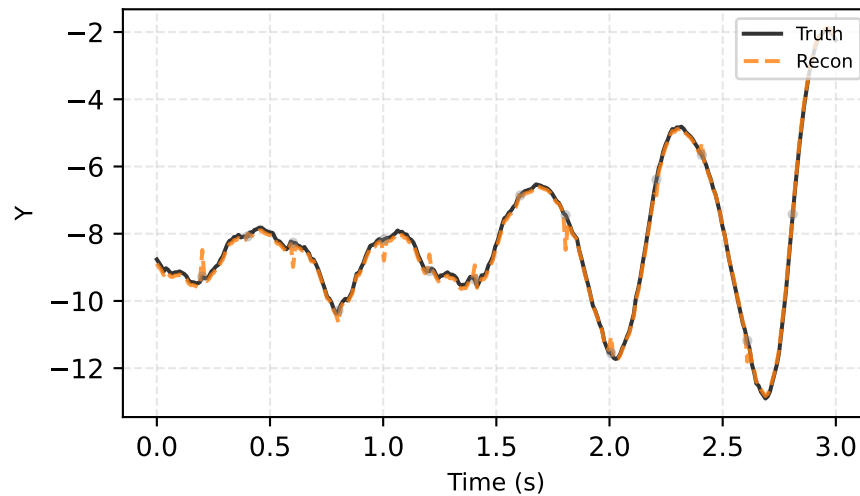


F1_vanilla_cfm_default (vanillaCFM) — CS2 Best & Worst Reconstruction

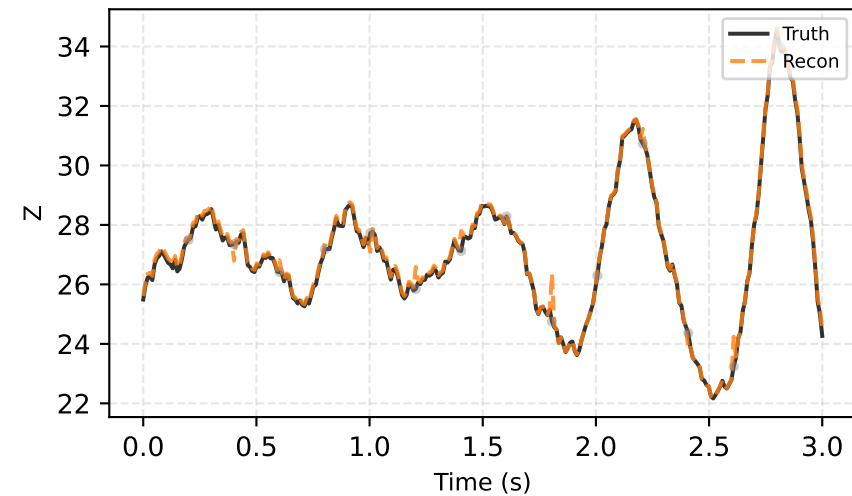
Best (traj #181) RMSE=0.129



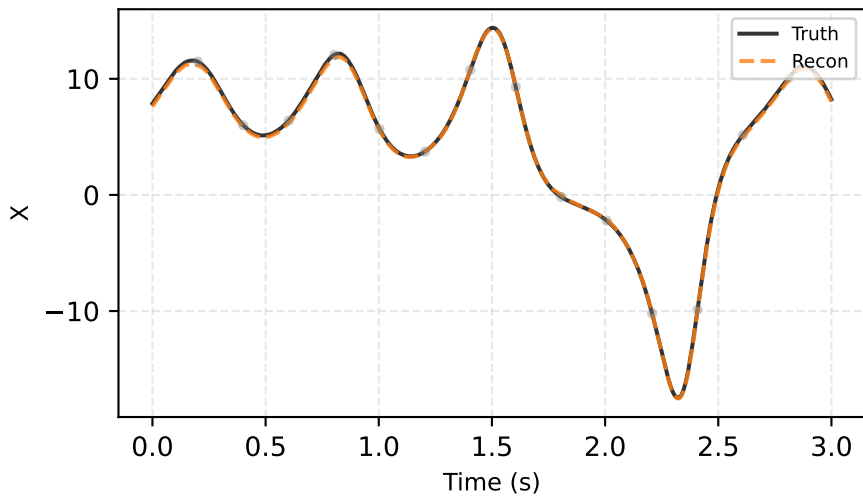
Best (traj #181) RMSE=0.129



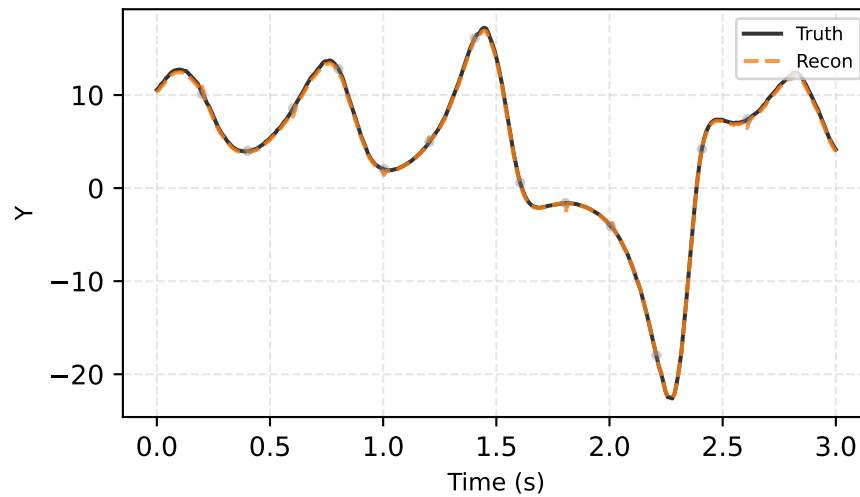
Best (traj #181) RMSE=0.129



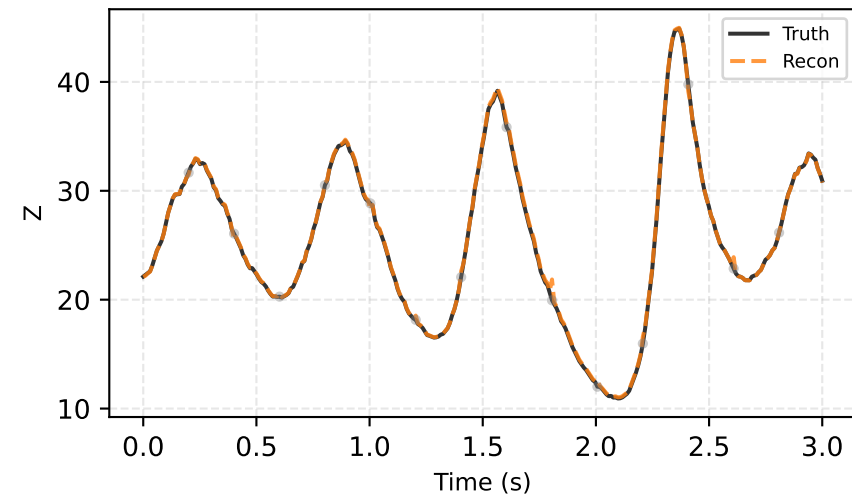
Worst (traj #114) RMSE=0.191



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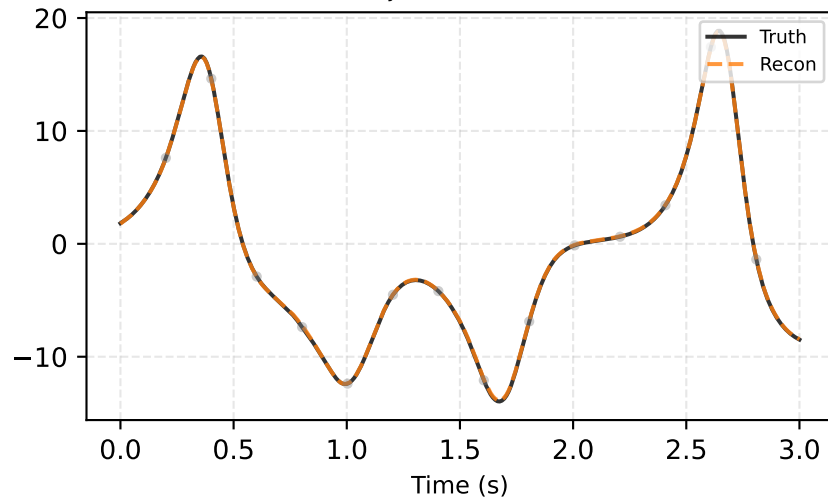


Worst (traj #114) RMSE=0.191

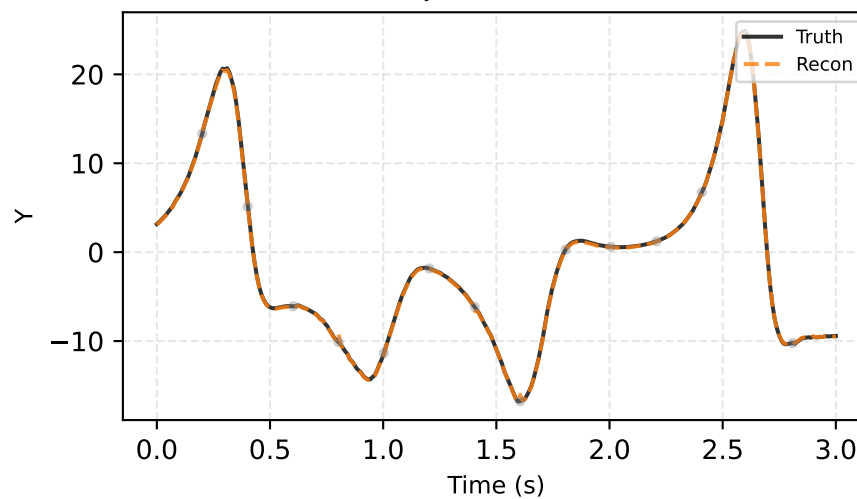


P2_vanilla_cfm_small (vanillaCFM) — CSI Best & Worst Reconstruction

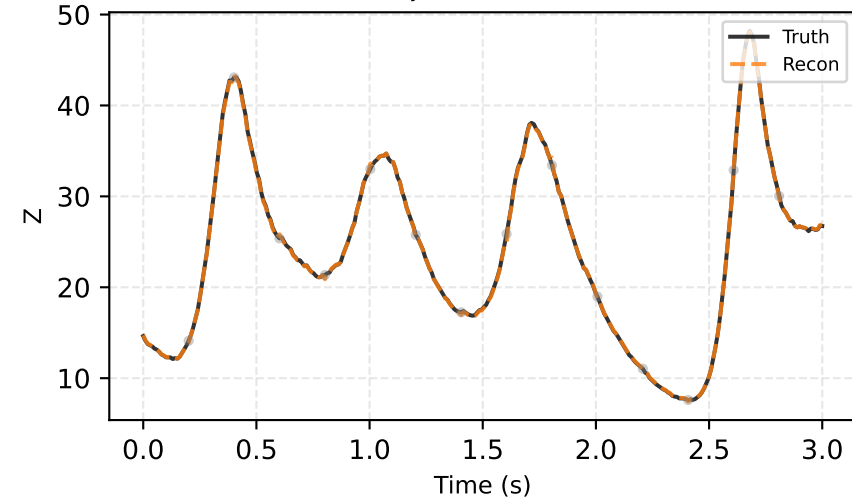
Best (traj #59) RMSE=0.105



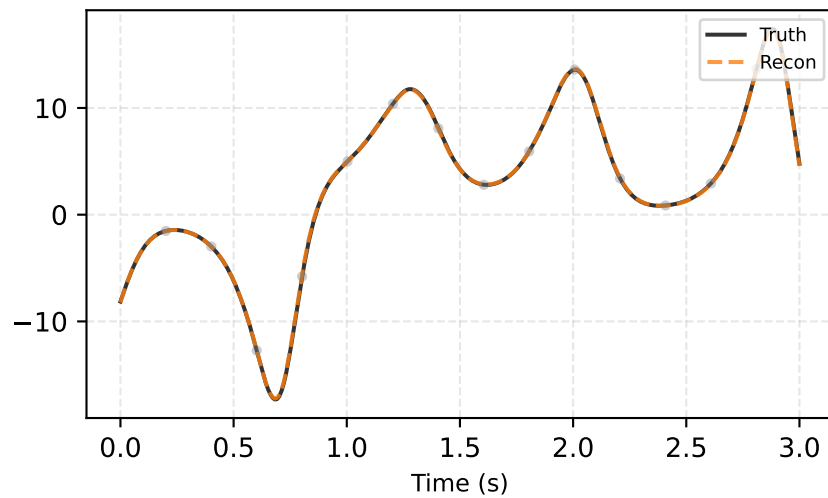
Best (traj #59) RMSE=0.105



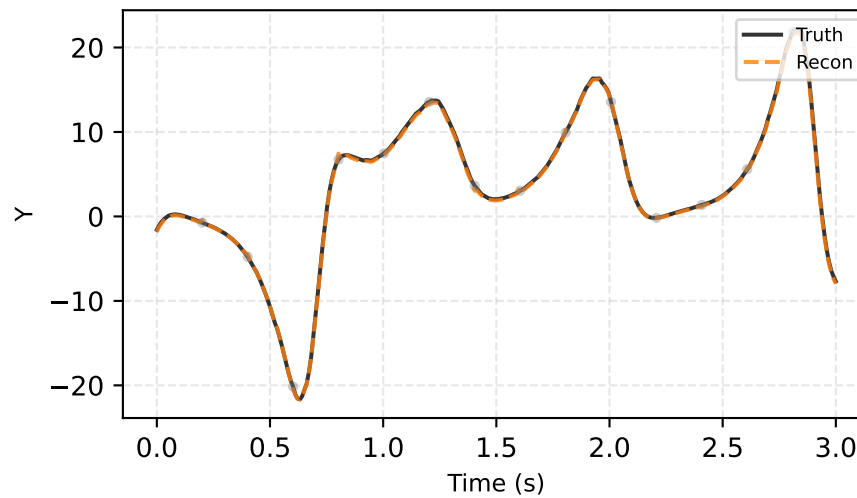
Best (traj #59) RMSE=0.105



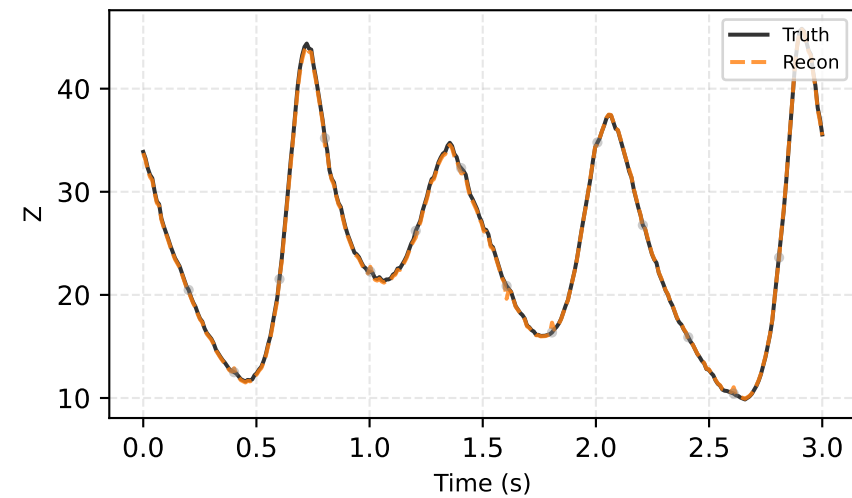
Worst (traj #86) RMSE=0.155



Worst (traj #86) RMSE=0.155

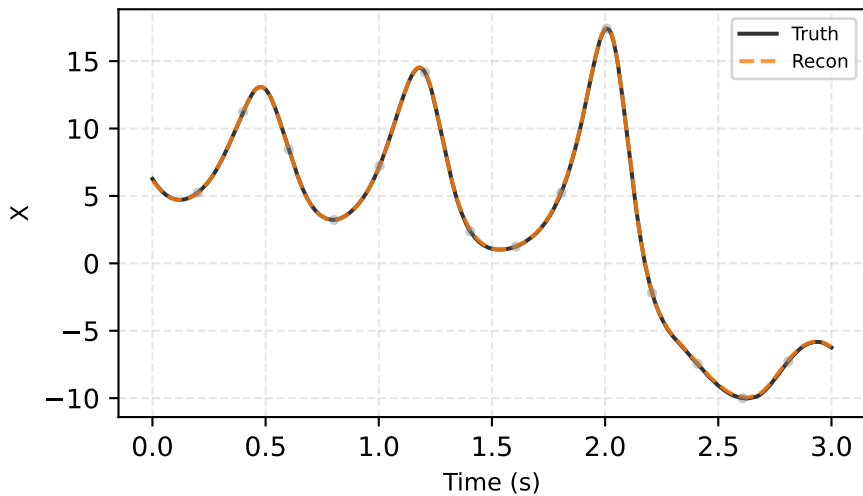


Worst (traj #86) RMSE=0.155

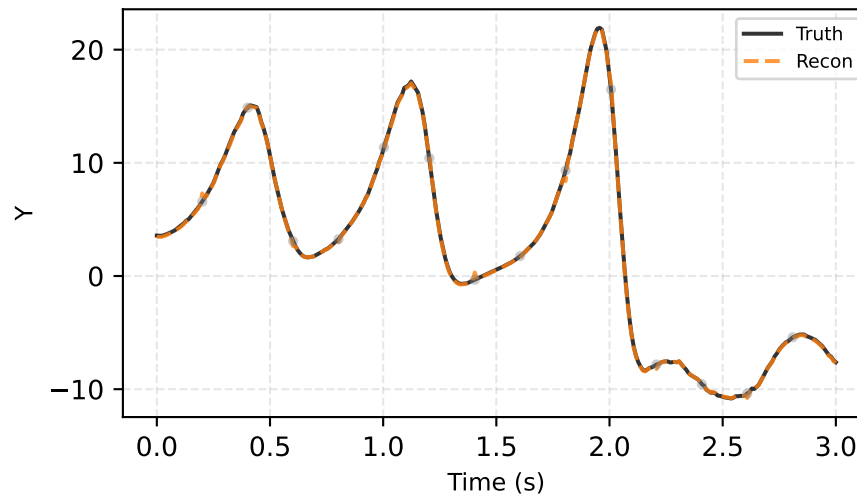


P2_vanilla_cfm_small (vanillaCFM) — CS2 Best & Worst Reconstruction

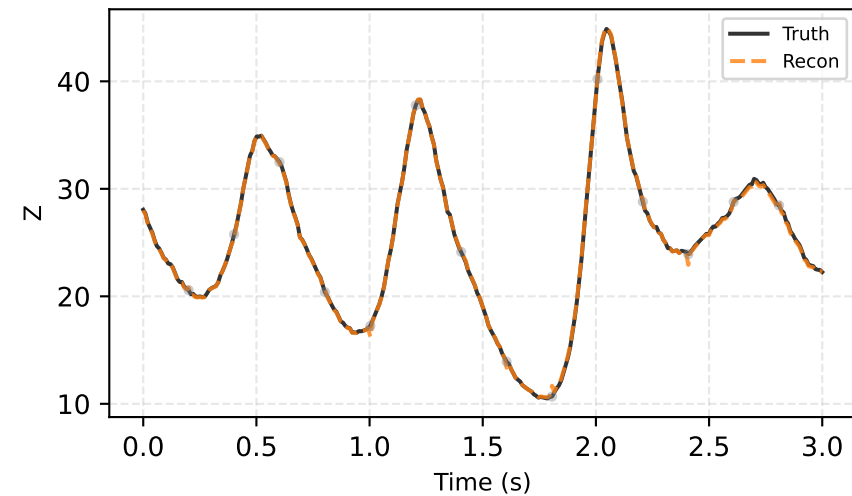
Best (traj #4) RMSE=0.115



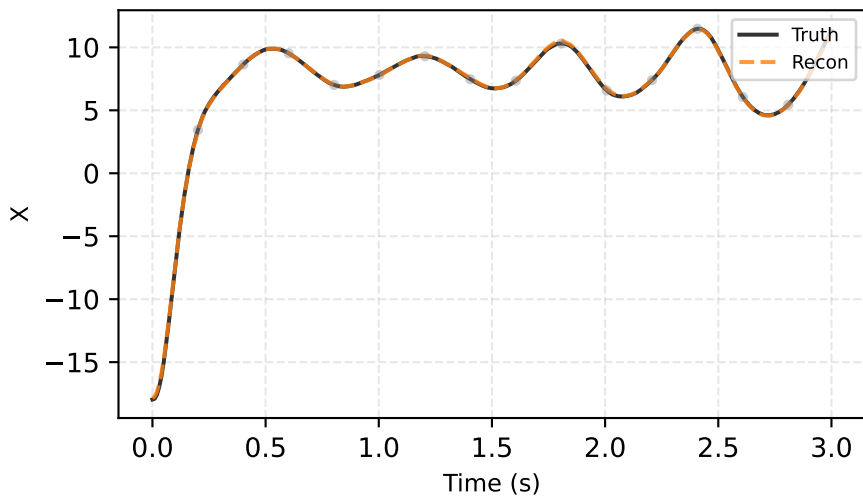
Best (traj #4) RMSE=0.115



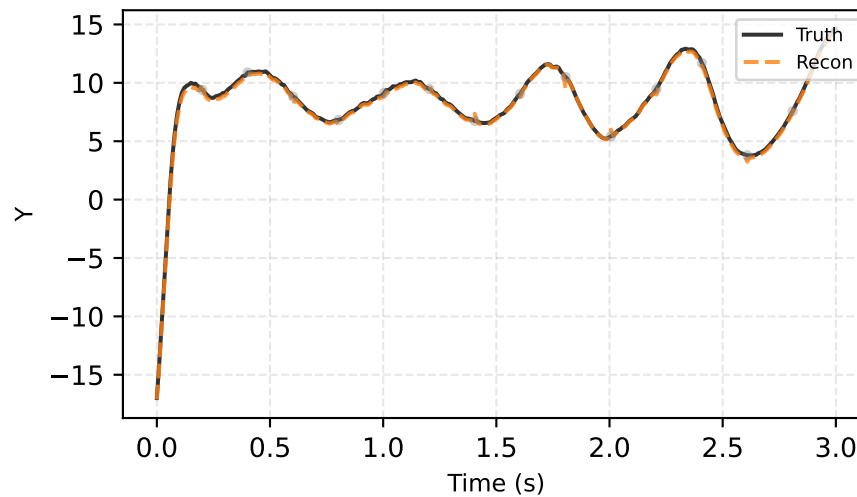
Best (traj #4) RMSE=0.115



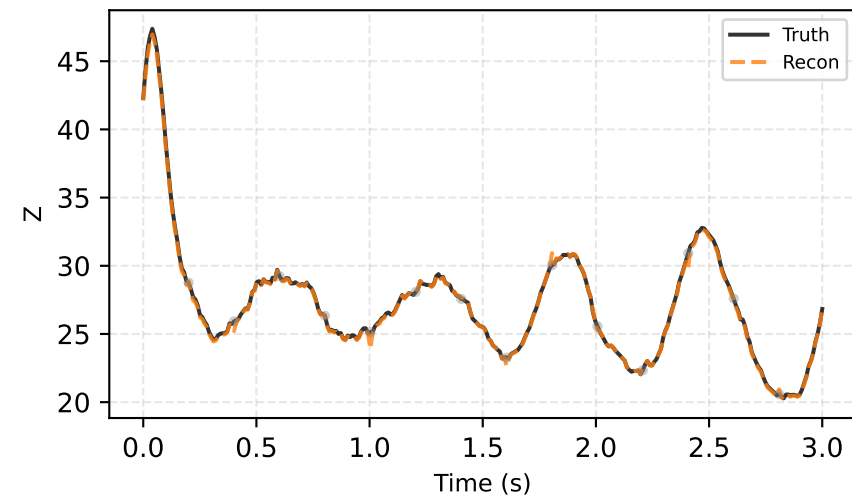
Worst (traj #127) RMSE=0.156



Worst (traj #127) RMSE=0.156

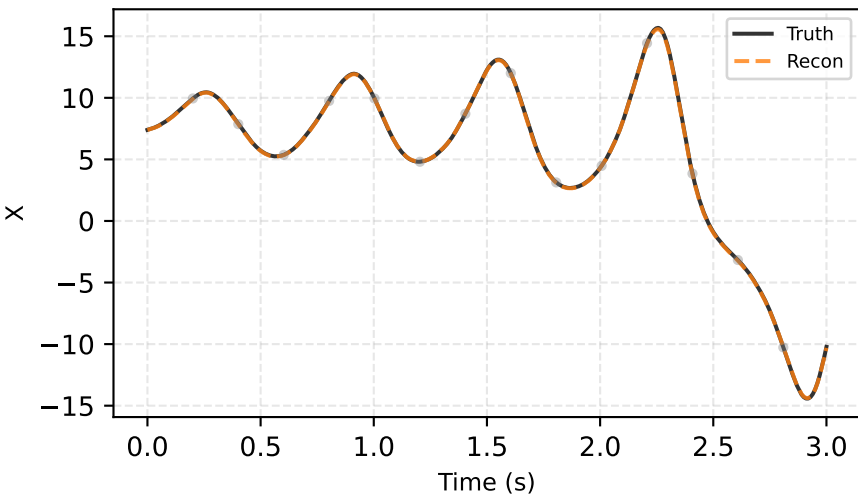


Worst (traj #127) RMSE=0.156

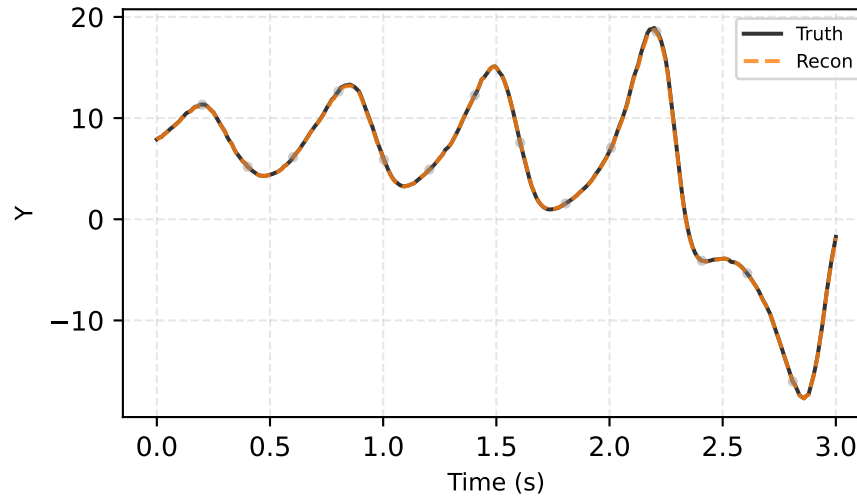


F5_vanilla_cfm_rand (vanillaCFM) — CS1 Best & worst Reconstruction

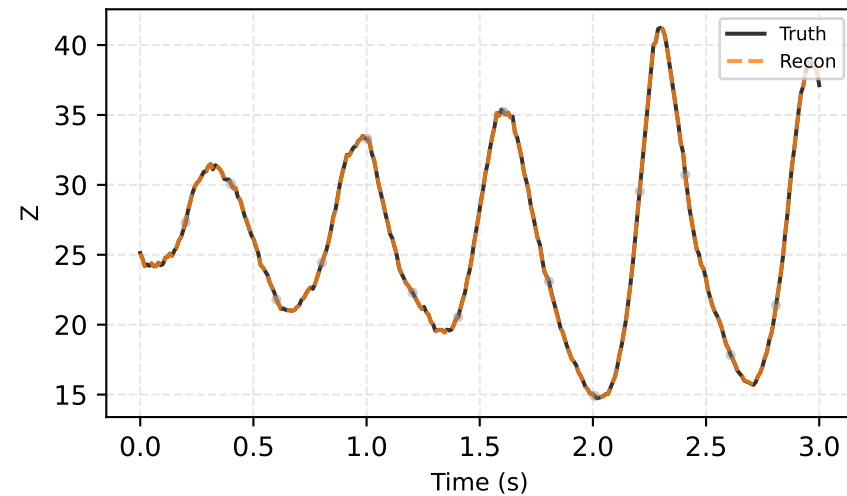
Best (traj #77) RMSE=0.051



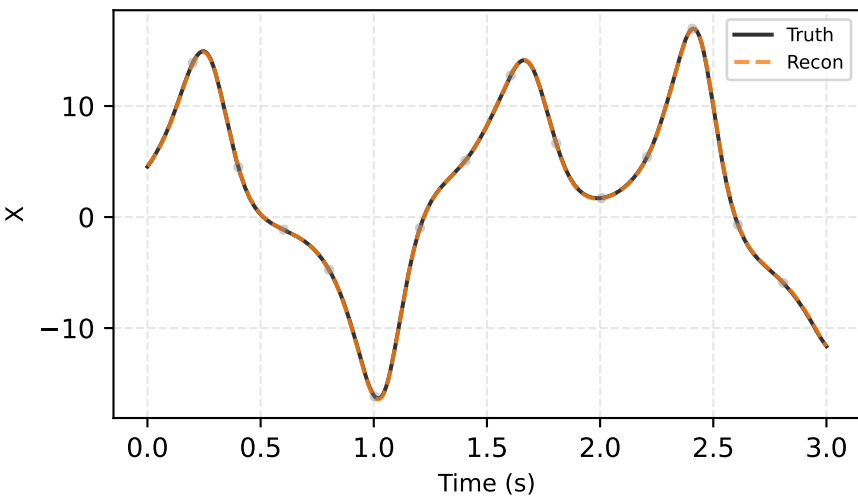
Best (traj #77) RMSE=0.051



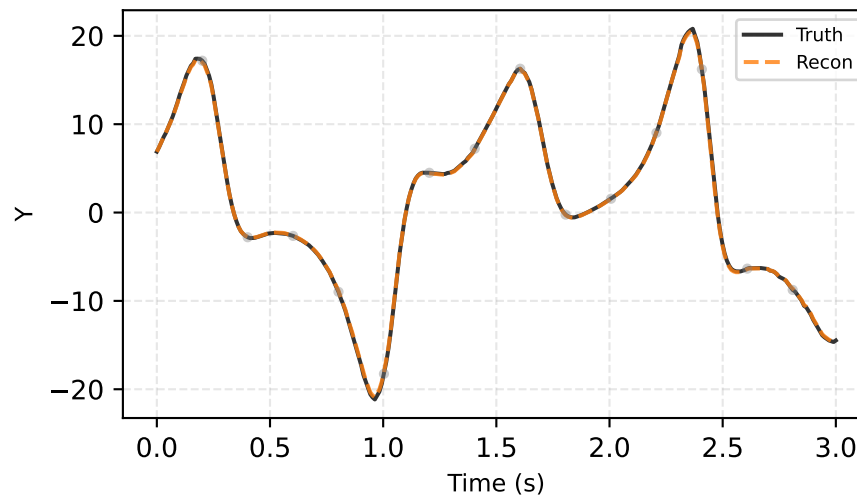
Best (traj #77) RMSE=0.051



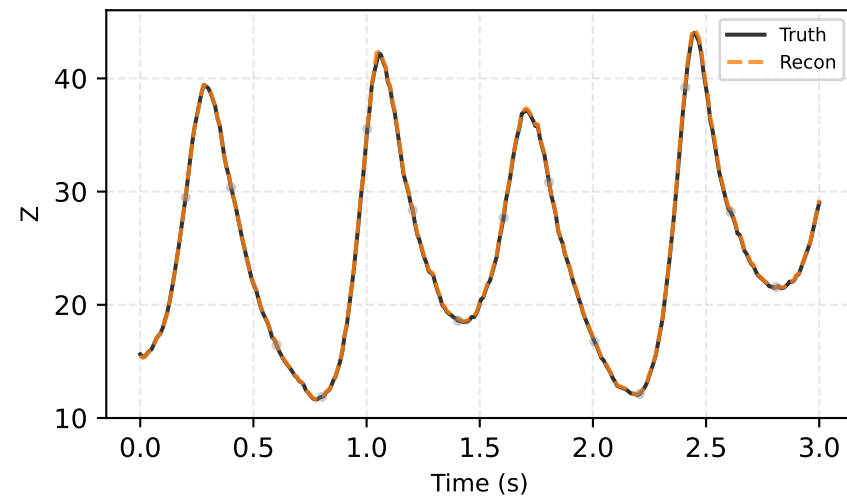
Worst (traj #100) RMSE=0.100



Worst (traj #100) RMSE=0.100

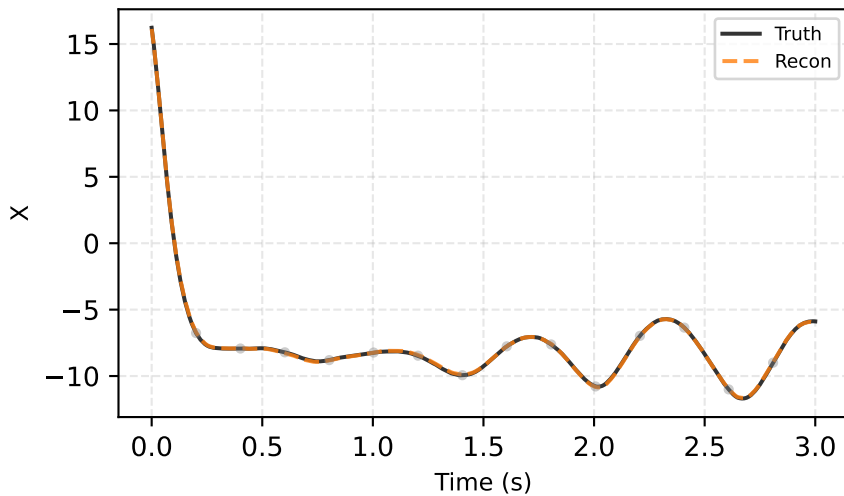


Worst (traj #100) RMSE=0.100

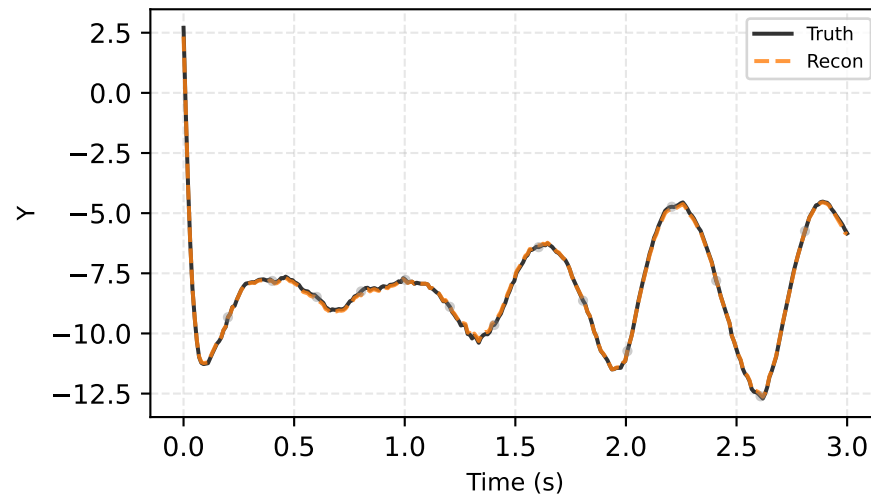


F5_vanilla_cfm_rand (vanillaCFM) — CS2 Best & Worst Reconstruction

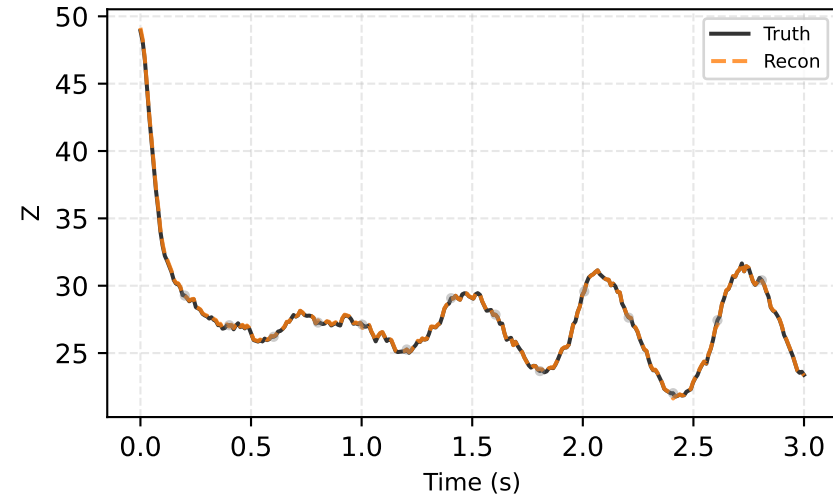
Best (traj #161) RMSE=0.051



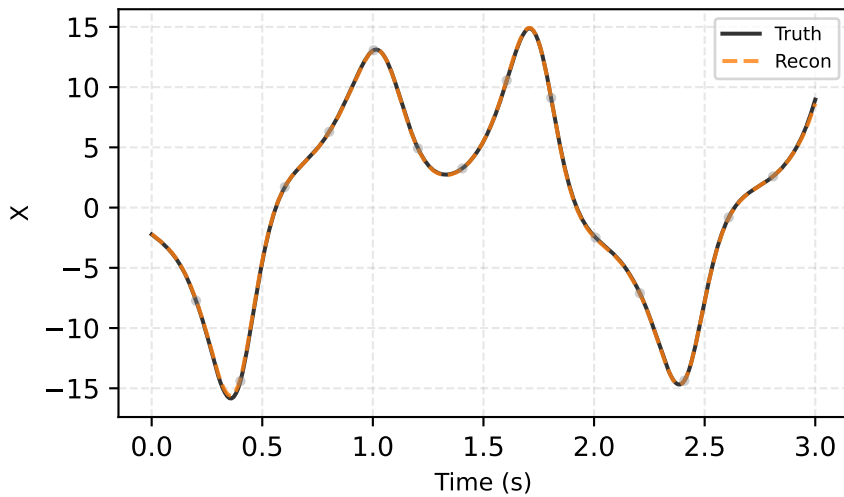
Best (traj #161) RMSE=0.051



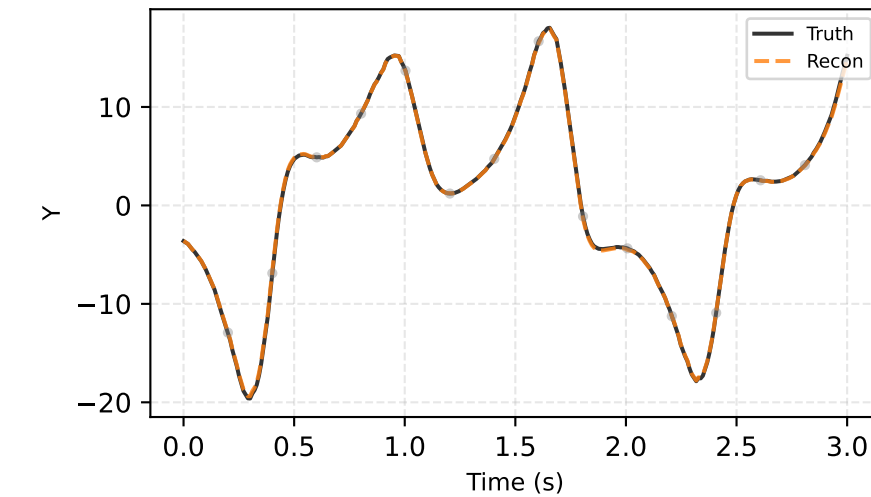
Best (traj #161) RMSE=0.051



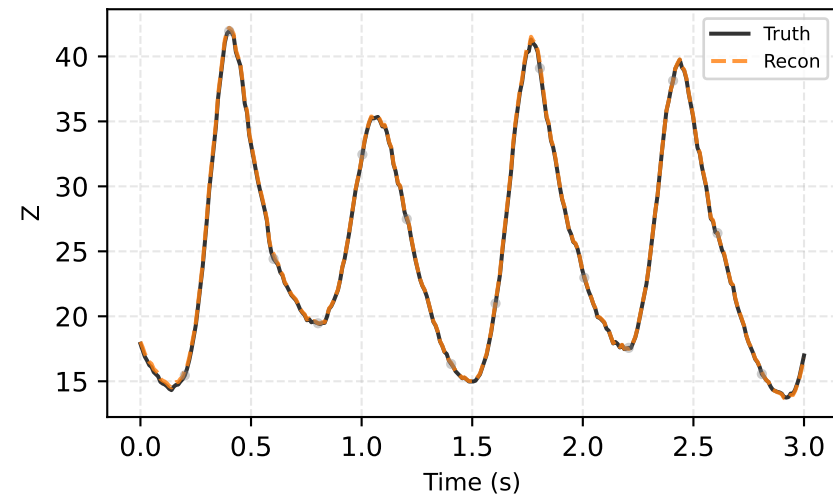
Worst (traj #44) RMSE=0.103



Worst (traj #44) RMSE=0.103



Worst (traj #44) RMSE=0.103



Conclusion

Conclusions & Recommendations

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Best Overall Model: F3 (VanillaCFM rand)

CS1 mean RMSE: 0.0690

CS2 mean RMSE: 0.0700

Degradation: 1.01x

Training time: 789s

Key Findings:

1. VanillaCFM with randomized parameter training (F3) is the overall best achieving near-perfect robustness (1.014x) and lowest absolute RMSE on both case studies (~0.07 on CS1, ~0.07 on CS2).
2. Small UNet architecture [32,64,128] consistently outperforms default [64,128,256] for both DirectUNet and VanillaCFM.
3. Randomized parameter training (param_noise=0.2) improves robustness across both model families.
4. DirectUNet achieves competitive results on CS1 (best: E2=0.0815) but degrades more significantly on CS2 (deg=1.252x).
5. VanillaCFM consistently achieves lower degradation (<1.06x vs >1.08x for DirectUNet on non-randomized training).
6. Per-variable: Z-component is hardest for DirectUNet (0.125 vs 0.087 for VanillaCFM). VanillaCFM handles all components more uniformly.

Best DirectUNet: E2 (small) – CS1=0.0815, CS2=0.1020

Recommendation:

Use F3_vanilla_cfm_rand for production deployment.

Consider further exploration of VanillaCFM with randomized parameters.